Leadership Plan

The MOSAIC leadership team has extensive experience directing large, multi-investigator research programs, interdisciplinary modeling efforts, and collaborative NIH-funded initiatives. Papin (PI of MOSAIC) has served as PI of an NIH T32 training program (with 35+ faculty mentors and 60+ trainees) for nearly ten years. He served as Director and PI of an NIH KL-2-funded faculty development program (with 40+ faculty across three institutions) for 5 years. In addition to other NIH, NSF, DOD grants, he is currently a co-lead of a project in an NIH U54-funded program. With his leadership as Editor-in-Chief of the journal PLOS Computational Biology and as an elected member of the Board of Directors of the Biomedical Engineering Society, he has leadership experience with large complex scientific organizations (100s of members of journal editorial board; 1000s of society members). Peirce-Cottler has been a member of a multi-institutional consortium funded by the Allen Foundation (with Agmon) on multiscale models of bacteria. Derré is currently working with an NIH U54-funded program for an “organ-on-a-chip” program for the female reproductive tract. Khetani is a part of a NIH UM1-funded consortium on New Approach Methodologies (NAMs) on chemical safety assessment. Agmon has worked with several NIH and DARPA funded multi-institutional programs on biomedical modeling. Young, Mac Gabhann, Arnold, and Kolling also have extensive experience with NIH U, P, and related grant programs.

There is a rich history of collaboration among team members. Papin and Mac Gabhann served as co-Editors-in-Chief of PLOS Computational Biology. Papin, Mac Gabhann, Peirce-Cottler, Arnold, and Agmon have an extensive collaboration history with multiple co-authored manuscripts and co-led grant applications among them. Papin and Derré have collaborated through academic advising roles. Khetani/Papin and Kolling/Young have long histories of scientific collaboration on grant and project development.

The Investigators will be organized as outlined in Figure 1. All Investigators, trainees, and staff will be a part of one of four scientific teams. Arnold will lead the BV team; Peirce-Cottler will lead the MASLD team; Mac Gabhann will lead the Integration team; and Agmon will lead the Software team. Each modeling-focused lab will have a lab member on each of the different teams. All experimental/clinical team members will be associated with the BV or MASLD groups according to their respective expertise. Team leads will report progress, challenges, and milestone attainment to Papin and Kolling during the monthly project meetings. These meetings will be used to coordinate activities across the BV, MASLD, Integration, and Software teams and ensure timely achievement of project milestones.

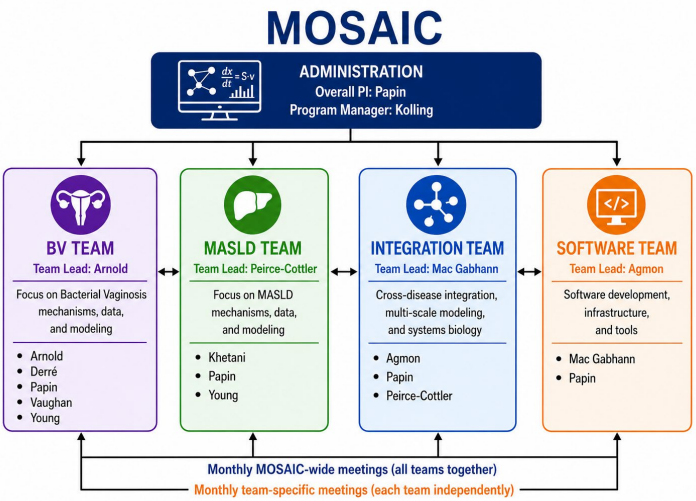


Figure 1. MOSAIC Leadership Structure

All project members will attend monthly online meetings to communicate project objectives, results and challenges. Project teams will meet separately on a monthly basis to discuss team-specific progress. All project members will communicate weekly as needed, either by Slack, phone, e-mail, or in-person to discuss smaller project-related matters. An annual in-person meeting will be held at the University of Virginia to present milestone accomplishments and discuss future objectives of the project among all team members. Kolling will serve as the program manager (with 50% effort committed) and will coordinate all communications.

The University of Virginia will establish sub-contracts with the respective institutions where each Investigator will be responsible for their own fiscal and research administration. Papin will provide overall scientific leadership for MOSAIC and will report all progress and necessary documents to the NIH, including progress reports and IRB approvals. Any scientific or administrative conflict will be addressed by the project leadership team and, if necessary, elevated to institutional leadership for resolution. If an Investigator moves to a new institution, attempts will be made to transfer the relevant portion of the grant to the new institution. In the event that an Investigator cannot carry out their duties, a new Investigator will be recruited as a replacement at one of the participating institutions.

Abstract

Sex-specific differences in therapeutic efficacy and toxicity are pervasive, yet the mechanistic role of dynamic hormone homeostasis in shaping these differences remains poorly understood. Current computational models fail to capture how variations in hormone regulation, metabolism, and host-microbiome interactions propagate across biological scales to alter drug performance. To bridge this critical gap, we will develop **MOSAIC** (**M**odeling **O**f **S**ex-specific metabolism **A**cross **I**nteracting **C**ommunities), an open-source, multi-scale computational framework designed to capture and predict hormonal, metabolic, and microbiome states across interacting organ systems. **MOSAIC** addresses the challenge of cross-scale physiological modeling by defining mechanistic interfaces that allow diverse, independently developed modeling paradigms to interact dynamically without sacrificing specific biological assumptions. A central focus is modeling microbiome-mediated estrogen metabolism and enterohepatic recycling as a systemic network linking gut, liver, and vaginal physiology. To ensure scalability and extensible curation, we will deploy innovative, AI-assisted tools to automate data harmonization, integrate existing multi-omic datasets, parameterize sex-specific metabolic networks, and prioritize simulation results for experimental validation. The key achievable goals will be tol: 1) Develop a multi-scale model of the hormone-regulated vaginal microbiome to predict local therapeutic interventions, 2) Build an integrated, sex-specific computational model of gut–liver host–microbiome metabolism incorporating hormone homeostasis to predict systemic therapeutic efficacy, and 3) Integrate the gut–liver and vaginal models to simulate system-level feedback, hormone-mediated transport, and drug-processing across compartments. Using bacterial vaginosis (BV) and metabolic dysfunction-associated steatotic liver disease (MASLD) as demonstration systems, **MOSAIC** will deliver the first human-relevant *in silico* platform capable of predicting therapeutic efficacy and adverse effects in hormone-regulated systems, ultimately accelerating precision medicine strategies.

Specific Aims

Evaluating how drugs and biologics perform across diverse patient populations is severely limited by our current inability to computationally map the systemic influence of sex hormones. While sex-specific variations in therapeutic efficacy and toxicity are widely acknowledged, the dynamic, system-level mechanisms driving these differences remain elusive. Existing metabolic models are restricted to single human cell types or localized microbial niches, failing to capture the physiological crosstalk linking host endocrine regulation, metabolic fluxes, and microbial ecology. Because perturbations in one compartment inevitably cascade across organ systems, **there is a critical need for a human-relevant, mechanistic framework** that treats hormone homeostasis as a dynamic regulator of host-microbiome and metabolic networks.

To bridge this critical gap, our overall objective is to develop, validate, and disseminate **MOSAIC** (**M**odeling **O**f **S**ex-specific metabolism **A**cross **I**nteracting **C**ommunities)**. MOSAIC** is an open-source, multi-scale computational framework that integrates host tissue metabolism, microbiome composition, and dynamic sex-hormone homeostasis to predict therapeutic efficacy and toxicity. Our **central hypothesis** is that explicit mechanistic interfaces between disparate modeling paradigms will allow independently developed models to function as an integrated representation of hormone-regulated human physiology. Our team is uniquely positioned to pioneer this approach using innovative modeling approaches with AI-based tools to automate data harmonization, integrate existing clinical multi-omic datasets, quantify parameter uncertainty, and prioritize high-value simulation results for experimental validation. We will demonstrate and validate this framework using publicly available datasets and *in vitro* approaches as needed to examine bacterial vaginosis (BV) and metabolic dysfunction-associated steatotic liver disease (MASLD) – two highly prevalent conditions with profound sex-specific components driven by host-microbiome-hormone-metabolism cross-talk – in the context of estrogen dynamics. The initial implementation of **MOSAIC** is focused on estrogen-mediated physiology in BV and MASLD as tractable demonstration systems. However, the framework is designed for extension to hormone-modulated diseases. Objectives will be achieved with the following aims:

Aim 1: Develop a multi-scale computational model of the hormone-regulated vaginal microbiome to predict local therapeutic efficacy. We will integrate ecological and metabolic models of vaginal microbial communities to assess how systemic metabolic and hormonal states shape local mucosal environments and community stability. Utilizing AI-assisted tools for model curation and parameterization, we will test the model's ability to predict the efficacy and adverse-effect risk of standard antibiotics and emerging biologics for BV across varying patient-specific hormonal states.

Aim 2: Construct an integrated, sex-specific model of gut–liver–microbiome metabolism under hormonal regulatory control. We will build a multi-compartment modeling framework that couples genome-scale representations of male and female hepatic metabolism with community metabolic models of the gut microbiome. The framework will be parameterized using human transcriptomic, metabolomic, and metagenomic datasets to capture hormone-regulated metabolic fluxes. We will validate the model using *in silico* and *in vitro* approaches of drug-processing pathways and therapies in the context of MASLD.

Aim 3: Integrate the gut–liver and vaginal models to predict system-wide, compartment-coupled therapeutic modulation. We will connect the models from Aims 1 and 2 to explicitly simulate the synthesis, microbial deconjugation, enterohepatic recycling, and systemic transport of sex hormones across biological compartments. We will predict how systemic variations in microbiome composition and endocrine status propagate across organ systems to alter downstream metabolic network behavior and patient-specific therapeutic responses.

Expected Outcomes and Impact: Upon completion, this work will deliver the first human-focused *in silico* framework capable of explicitly modeling hormone homeostasis as a dynamic, system-level regulator of host-microbiome interactions. By providing an open, modular, and reusable platform, MOSAIC will transform how the scientific community investigates sex-specific physiology, driving the accelerated design of precision therapeutic strategies.

Public Health Relevance

Tens of millions of women per year in the United States are affected by bacterial vaginosis and metabolic dysfunction-associated steatotic liver disease (MASLD), two conditions strongly influenced by dynamic hormone regulation and interconnected metabolic and microbiome processes. Progress has been hindered by limited understanding of the complex interplay among hormonal, physiological, metabolic, and microbiome systems. This program will develop mechanistic computational models to uncover the biological mechanisms underlying these interactions and to predict sex-specific therapeutic efficacy and toxicity, ultimately enabling improved therapeutic strategies for women’s health.

MOSAIC: A mechanistic, multi-scale modeling framework for hormone-regulated host-microbiome metabolism to predict sex-specific therapeutic efficacy and toxicity

MOSAIC (**M**odeling **O**f **S**ex-specific metabolism **A**cross **I**nteracting **C**ommunities) addresses a fundamental barrier in computational modeling for clinical applications: sex-specific differences in therapeutic response often arise from interactions across biological systems that are still modeled in isolation. Hormones, metabolism, microbial communities, tissue physiology, immune state, and drug exposure form a coupled physiological network, but current models rarely capture how changes in one subsystem propagate through the others to alter efficacy or toxicity. We will develop an open, modular modeling framework that connects these mechanisms across scales while preserving the assumptions, data sources, and biological scope of each component model. By making these interactions executable, shareable, and reusable, **MOSAIC** will enable mechanistic predictions of how hormone-regulated host-microbiome dynamics shape therapeutic response in bacterial vaginosis (BV), metabolic dysfunction-associated steatotic liver disease (MASLD), and future disease contexts where sex-specific physiology remains an unresolved driver of clinical variability.

Individuals with similar physiological measurements can experience markedly different clinical outcomes. We focus on two conditions for which the complex interplay between hormone homeostasis, metabolism, and the microbiome manifests in sex-specific ways and for which responses to therapeutic interventions are difficult to predict. First, BV affects more than 20 million women in the US (1). It arises from dysbiosis of the vaginal microbiome where the interaction between hormone dynamics and host environment results in enormous challenges to resolve the condition (2). Second, MASLD affects nearly 100 million people in the US and nearly half of affected individuals are women, with marked differences between pre- and post-menopausal populations (3,4). MASLD arises from a complex interplay between liver metabolism, gut microbial ecology, immune responses, and hormone regulation, with perturbations in one subsystem propagating across scales to influence disease progression (5). Treatment strategies for these two conditions can vary widely in efficacy. For example, two women receiving the same treatment for BV may experience different recurrence patterns due to interactions among vaginal microbial ecology, estrogen-regulated epithelial physiology, immune responses, metabolic profiles, and antibiotic exposure (6). Existing computational models are typically designed to address individual components of these systems—for example, predicting drug exposure, microbial community dynamics, or metabolic activity—but they rarely capture how these processes interact across biological scales. As a result, important outcomes, including therapeutic efficacy and toxicity, may emerge not from any single subsystem but from feedback among hormones, metabolism, microbial communities, immune responses, and therapeutic interventions. **MOSAIC** is designed to make these interactions executable, enabling mechanistic explanations and predictions of heterogeneous responses to therapeutics in hormone-regulated physiology. The breakthrough is not any single component model, but the executable, hormone-centered coupling layer that allows independently validated metabolic, microbiome, PKPD, and agent-based models to propagate sex-specific therapeutic effects across tissues.

Scientific rationale and justification of the approach

Sex-specific differences in therapeutic efficacy and toxicity are pervasive, yet the mechanistic role of dynamic hormone homeostasis in shaping these differences remains poorly understood. Current computational models do not capture variability linked to hormone regulation of metabolism and host-microbiome interactions, limiting predictions on how drugs and biologics perform across sex-specific physiological states. To address this gap, we will develop a computational framework to capture hormonal, metabolic, and microbiome states in multi-scale computational models that predict therapeutic efficacy and toxicity. This framework uniquely integrates genome-scale metabolic models of human and microbial physiology (Papin), dynamic microbiome models (Arnold), multicellular agent-based models (Peirce-Cottler), whole-body compartment models (Mac Gabhann), and robust software design of integrative modeling frameworks (Agmon), enabling mechanistic assessment of the role of metabolism, microbial species, and sex hormones in therapeutic efficacy. These models may also resolve open questions about the causal drivers vs. correlative factors associated with sex-specific differences in disease. We will implement this framework as an open-source platform termed **MOSAIC**. We will leverage team expertise in all these modeling methods as well as domain-specific expertise in hepatic (Khetani) and vaginal (Derré) cell biology, along with clinical expertise in the gastrointestinal (Young) and vaginal (Vaughan) microbiome.

As an initial proof-of-concept, we will focus on BV, a condition in which hormone-regulated shifts in the vaginal microbiome and metabolic environment directly influence therapeutic response to antibiotics and

emerging biologics (probiotics). We will extend the framework to MASLD, where gut microbiome-derived metabolites and hormone-regulated hepatic metabolism influence sex-specific therapeutic efficacy and toxicity, with reported sex-specific differences in efficacy of existing drugs including GLP-1 receptor agonists as examples (7,8). Despite both BV and MASLD being at the interface of metabolism and the microbiome, these prevalent conditions with clear sex-specific differences are typically studied in isolation.

A key driver linking metabolism and the microbiome, particularly in the context of these diseases, is systemic hormone homeostasis (9,10). For example, the gut microbiome contributes to systemic estrogen availability through microbial deconjugation and enterohepatic recycling, while estrogen modulates hepatic metabolism, immune responses, and vaginal epithelial physiology (11–14). Perturbations in the dynamics of estrogen and other sex-specific hormones—arising from microbiome composition, body composition, dietary inputs, or reproductive stage—have the potential to propagate across organ systems. These interactions suggest that hormone-regulated metabolism represents a unifying lens to examine seemingly disparate disease states and dysbiotic conditions. Although the initial implementation focuses on estrogen biology because of its central role in BV and MASLD, the framework is designed to accommodate additional endocrine signals including androgen, thyroid, glucocorticoid, and peptide hormone signaling. BV and MASLD are used as demonstration systems because both exhibit strong dependence on hormone-regulated physiology. The primary objective of **MOSAIC** is not disease-specific prediction, but rather the development of a reusable framework for understanding how perturbations in hormone homeostasis propagate across biological scales to influence therapeutic response.

MOSAIC will enable identification of potential therapies for metabolic and microbial targets and prediction of adverse effects in a sex-specific manner. It will also allow for the integrated modeling of the gut, liver, and vaginal systems and explicitly represent synthesis of sex-specific hormones, microbial metabolism, transport, and enterohepatic recycling, enabling simulation of hormone-mediated feedback across physiological compartments. This model will output predictions about the roles of liver and microbiome metabolism in processing other drugs. Sex-specific physiology will be incorporated through differential parameterization of metabolic networks and hormone dynamics, enabling patient-specific models (digital twins).

Most existing metabolic and dynamic models focus on individual cell- or tissue-types or isolated microbial species. Rarely do models incorporate dynamic hormone regulation or sex-specific physiology. As a result, there is no human-relevant, mechanistic framework for exploring how microbiome-mediated changes in hormone homeostasis influence metabolic fluxes, microbial ecosystem stability, or sex-specific responses to therapeutic interventions. **Our approach represents the first attempt to explicitly model hormone homeostasis as a dynamic, system-level regulator of host-microbiome metabolic interactions across multiple organ systems (Figure 1).** These models will require innovation in AI-based tools for model curation and parameterization, multi-scale model coupling, and data integration to capture sex-specific human physiology.

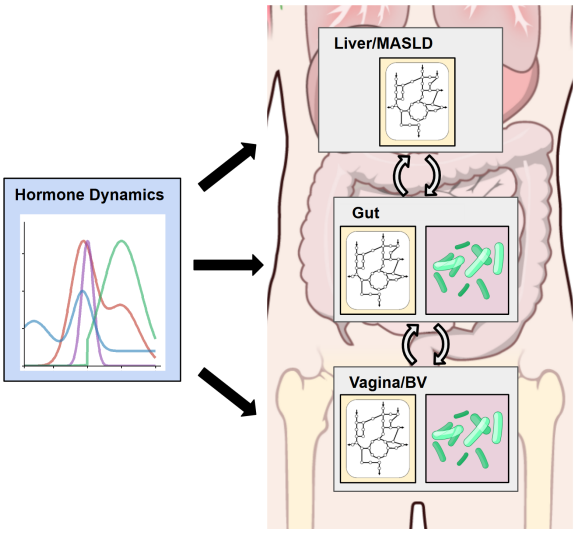


Figure 1. MOSAIC makes predictions across three tissue types: the liver, the gut, and the vagina. Each tissue type will be modeled by a genome-scale metabolic model, and the gut and vaginal tissues will include additional relevant genome-scale metabolic models, mechanistic community models, and agent-based models according to their specific microbial community. Together, these compartments will be used to study how MASLD in the liver and BV in the vagina affect metabolism and microbiome composition in these tissues.

To enable scalable and extensible modeling, **MOSAIC** will incorporate AI-based tools to integrate and parameterize existing multi-omic datasets (some of which were recently developed in our own labs), automate data harmonization, quantify uncertainty, and facilitate model curation. Our labs are pioneering the application of AI-based tools to mechanistic computational models. AI-based tools will also enable the identification of the most informative simulation results to prioritize experimental validation efforts. This work will deliver an innovative, human-focused *in silico* framework capable of supporting hypothesis generation, model refinement, and broad community use as a reusable platform to study hormone-regulated host-microbiome interactions.

Short-term (first year) objectives

- Curate existing data for MASLD and BV model construction
- Implement AI-based data curation, integration, and parameterization tools
- Construct a hormone-regulated vaginal microbiome digital twin model
- Establish model credibility criteria and validation pipeline

Long-term (second and third year) objectives

- Construct a sex-specific, hormone-regulated gut-liver-microbiome digital twin model
- Integrate vaginal and gut-liver modules into a unified **MOSAIC** platform
- Enable patient-specific, sex-specific therapeutic predictions
- Identify mechanistic therapeutic targets and prioritize experimental validation
- Foster community adoption and extensibility

Description of the proposed model

MOSAIC provides a mechanistic, quantitative approach to understand how individual microbial species and community-wide metabolic networks collectively respond to sex-specific hormone levels and dictate therapeutic responses. A core innovation of our framework lies in its automated, individualized pipeline. Each digital twin produced by **MOSAIC** will provide an accurate reflection of the real-time ecological forces shaping an individual's microbiome environment under varying lifestyle factors and sex-specific transcriptomics. Ultimately, we will leverage this digital twin framework to 1) predict patient-specific therapeutic efficacy and adverse drug reactions before treatment administration, 2) design precise nutritional, biotherapeutic, or hormonal interventions that restore homeostatic balance and optimize drug response, and 3) elucidate sex-specific differences in the dynamic bidirectional networks driving these responses. By providing a rigorous, automated, and scalable mathematical foundation for digital twins, this work shifts the paradigm from descriptive association studies to predictive, personalized medicine—paving the way for targeted therapies that account for sex-specific differences and optimize both endocrine health and drug efficacy. The integrative framework for **MOSAIC** is outlined below:

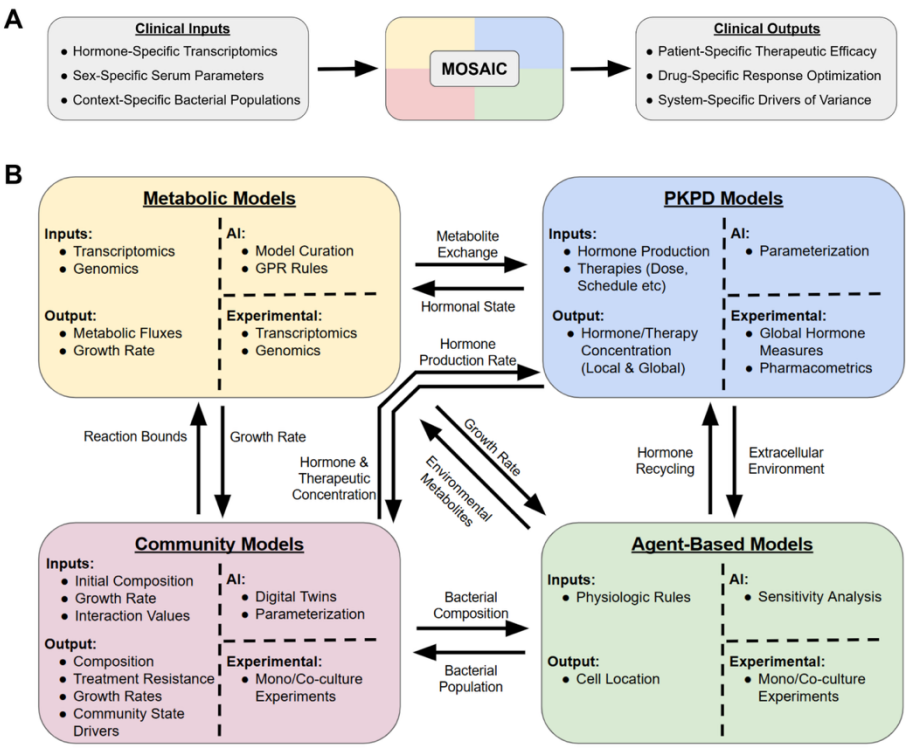


Figure 2. (A) MOSAIC can use several relevant clinical inputs, such as hormone-specific transcriptomics, sex-specific serum-derived parameters, and context-specific bacterial populations, and produce many important clinical outputs, such as hormone-specific therapeutic effects, sex-specific metabolic changes, and context-specific microbiome population shifts. (B) Each modeling approach has relevant inputs, outputs, AI usage, and in vitro experiments. Importantly, outputs from one approach can serve as a way to improve assumptions and inputs to other modeling approaches, meaning changes to one approach propagate out to influence other modeling approaches.)

*Context-specific genome-scale **metabolic models***

We will create genome-scale metabolic models (GEMs) for each human tissue and microbiome community in a high-estrogen and low-estrogen environment. The liver and gut models will be created from Human1 , a previously published general human cell GEM, and publicly available transcriptomic data (see data resources section below) in combination with fast task-driven integrative network inference for tissues (ftINIT) (15), which will inform which reactions are most important for the given tissue type and hormone environment. We will use transcriptomic data for the microbial

species in environments that simulate the associated physiological niches (gut, vagina), with particular attention to *Gardnerella* and *Lactobacillus* species for the vaginal microbiome (16) and *Klebsiella*, *Escherichia*, *Streptococcus*, *Faecalibacterium*, and *Roseburia* species for the gut microbiome (17) given their known associations with BV and MASLD, respectively. As needed, we can also generate associated transcriptomics data with *in vitro* conditions that simulate these physiological environments. Metabolic network reconstructions are readily generated at these scales with the software tool Reconstructor (18) which we developed. LLMs will be used to curate gene-protein-reaction rules in the GEMs to improve the prediction of gene perturbations and drug targets with a pipeline we recently developed. We will sample the feasible flux distributions in these tissue- and environment-specific models to predict metabolic reaction activity and growth rates. AI-based random forest or LASSO tools will be used to select a sparse set of discriminatory reactions from high-dimensional flux datasets and identify those most predictive of different physiological states. Papin's team has used these methods previously for human models to successfully predict inflammatory metabolites in Crohn's disease (19) and targetable genes in gastrointestinal bacteria (20).

Mechanistic microbiome ***community models***

Existing computational approaches to study the microbiome are siloed. Ordinary Differential Equation (ODE) models excel at capturing the macro-level, population-level dynamics of microbial communities over time, but manual parameterization for individual patients is bottlenecked by data availability and species selection, and internal cellular metabolism is treated as a "black box". Conversely, Genome-Scale Metabolic Models (GEMs) provide high-resolution, constraint-based reconstructions of intracellular metabolic pathways, but they struggle to inherently capture time-varying population shifts and complex intercellular dependencies in a dynamic ecosystem. This proposal addresses these critical methodological gaps by introducing an innovative, multiscale computational framework designed to construct microbiome digital twins. Arnold's team has pioneered the development of microbiome community models for the prediction of microbial dynamics in BV (21–23). This proposed approach utilizes data-driven models to capture personalized species composition and parameterize dynamic ODEs, subsequently isolating and prioritizing key microbial interactions for high-resolution metabolic evaluation using Genome-Scale Metabolic Models (GEMs). This framework bridges the gap between macro-scale ecosystem dynamics and micro-scale metabolic fluxes to relate host hormone bioavailability to drug efficacy with unprecedented precision.

Pharmacokinetic/pharmacodynamic ***(PKPD) models***

We will construct a multi-compartment whole-body pharmacokinetic (PK) model that includes compartments representing the key organs for BV and MASLD (e.g., gut, liver, kidney, female reproductive tract), connected via a central blood compartment, to track systemic and local levels of key sex hormones and therapeutics. The PK model will mechanistically represent estrogen biosynthesis, therapeutic delivery dosage (including route, dose, and time), tissue distribution, and hepatic/renal clearance, parameterized with sex-specific physiological data. Each molecular species is tracked by one equation in each compartment, and the terms in these equations are the quantitative rates representing each biological mechanism that affects that molecule. Mac Gabhann has pioneered such approaches for sex-specific prediction of cell proliferation in the female reproductive tract (24). Drug-based therapies, such as antibiotics (e.g., metronidazole) for BV or resmetirom for MASLD, will be modeled with oral delivery, where both the local and systemic effects following absorption will be modeled. Non-drug therapies, such as biologics (probiotics like *L. crispatus*) for BV, will be modeled with delivery of the probiotics into the community model, from which the resulting modified production of hormones will be an input to the PK model. The pharmacodynamic (PD) component will couple predicted plasma estrogen concentrations to estrogen receptor α/β and therapy-specific binding kinetics in target tissues, plus the impacts of exogenous drugs on the hormone system. The PKPD model takes as input the hormone production levels from the community and agent-based models and therapeutic dosing based on the specific clinical interventions being modeled. The outputs of the PKPD model include systemic and local hormone and drug levels, which serve as inputs to the community model, agent-based model, and metabolic model, quantifying the hormone and drug environment in which each of those models are set. Outputs from this model can be used to inform what kind of hormonal environment would be most appropriate to use in our other models. Experimental measurements of the plasma levels of hormones and drugs are measured as possible so that the predictions of the model can be compared to the clinical pharmacometrics. This direct clinical connection facilitates the parameterization of the models using sex-specific data. Mac Gabhann's team has

created multiple similar PKPD models that simulate not only exogenous therapeutics including biologics such as antibodies (25,26), gene therapies (27), and cell therapies (28,29), but also simultaneously tracking the relevant endogenous molecules that are distributed systemically such as hormones and cytokines (24,30,31), and the PD interactions between the exogenous and endogenous molecules.

Agent-based models

We will implement an agent-based model (ABM) in NetLogo (32) to simulate the spatiotemporal dynamics of bacterial-epithelial interactions in the gut and vaginal microenvironments. Individual bacterial agents and epithelial cells will be assigned stochastic, rule-based behaviors, including adhesion, proliferation, apoptosis, and cytokine secretion, governed by local environmental conditions. Hormonal concentrations (estrogen and progesterone) will be incorporated as diffusible extracellular fields that modulate agent behavior. Machine learning and AI will be applied to optimize agent rules and identify key cellular interactions driving sex-hormone-dependent dysbiosis or homeostasis. Peirce-Cottler framed the importance of modeling complex biological systems, including immune responses (33,34), in terms of interacting agents and emergent properties, establishing a foundation for multiscale modeling and hybrid computational approaches (35). Methodologically, the development of tools such as Vivarium (a collaboration between Peirce-Cottler and Agmon) has enabled flexible integration of different types of modeling approaches, including agent-based components, facilitating reusable, composable simulations of biological systems across scales (36). These tools and approaches have been deployed in recent research applications, such as whole-cell/agent-based simulations of *E. coli* colonies that quantify single-cell heterogeneity in antibiotic responses (37) and spatial transcriptomics-driven multiscale models of *P. aeruginosa* biofilms (a collaboration between Peirce-Cottler and Papin) that connect metabolic states to microenvironmental stressors (38).

Model Integration

Hormone homeostasis provides the principal coupling mechanism among MOSAIC modules, linking microbial metabolism, tissue physiology, and therapeutic exposure across scales. The goal of **MOSAIC** is to leverage the multitude and diversity of outputs from one modeling method as inputs or constraints for the other modeling methods. These modeling interactions, which are summarized in Figure 2, are as follows:

- Metabolic models provide flux samples and growth rates as output. Growth rates can be used to parameterize individual bacterial species or different host cell types in a given context in both mechanistic community models and agent-based models.
- PKPD models provide therapeutic and hormone concentrations as output. These concentrations can be used to inform the hormone state of the metabolic models and as starting concentrations for the mechanistic community models and agent-based models.
- Mechanistic community models provide community composition, treatment resistance, growth rates, and community state drivers as outputs. Community composition can influence flux bounds in metabolic models, hormone production rate in PKPD models, and agent population number in agent-based models.
- Agent-based models provide cell location and agent behavior as output. These outputs can then parameterize key features of the community models and inform how metabolic models exchange metabolites.

Together, these interactions will help predict the efficacy of several therapeutic strategies. As an example for BV, we can include different biologic (probiotic) or pharmaceutical treatments, such as *Lactobacillus crispatus* or metronidazole, in the **MOSAIC** model. These interventions will modulate metabolite exchange in the metabolic models, growth rates in the community models, hormone production rates in the PKPD models, and cell populations in the agent-based models. Species can be ranked according to their effect toward an anti-BV environment in a given hormonal context. As an example for MASLD, we can predict how a drug such as resmetirom or GLP-1 receptor agonists affects liver metabolism in high- and low-estrogen environments. PKPD modeling would describe how the drug's concentration would change over time, thus determining metabolic pathway changes in the metabolic models. These changes would propagate out to the community and agent-based models via a difference in metabolites exported from the metabolic models. The final prediction would be a decrease in lipid metabolism specific to hormonal context.

AI-empowered modeling tools

AI-driven methods are integral to the **MOSAIC** framework and will support model parameterization, rule curation, and analysis of high-dimensional simulation outputs. AI tools will ultimately improve physiological accuracy and enable identification of key biological drivers underlying a highly complex modeling network. For example, in the personalized microbiome community models, we will use an AI-based adaptive parameterization architecture that can be adjusted based on clinical data availability. When longitudinal clinical data is available, we will use a deep personalization approach by employing a novel AI Autoencoder and Neural Ordinary Differential Equation (Neural ODE) pipeline. A variational autoencoder (VAE) will first compress high-dimensional, noisy clinical time-series data into a low-dimensional latent space representing the underlying ecological states and Lotka-Volterra (LV) model parameters. A Neural ODE network will then learn the continuous-time dynamics within this latent space. By passing these learned gradients back into our mechanistic generalized LV framework, we can efficiently infer high-confidence, individualized interaction parameters and growth rates tailored precisely to that individual patient's *in vivo* physiology. These parameters learned in one data set can also be used as parameter range estimates for other data sets where longitudinal data is not available. Alternatively, when longitudinal clinical data is absent from the beginning, we will use alternative parameterization paths to either create *in silico* patient cohorts through Latin Hypercube Sampling (LHS), as done previously (39), or use empirical scaling of parameters in the LV framework based on published *in vitro* co-culture experiments that inform interaction terms (40). These alternative approaches ensure our model can perform broad predictive screening of drug efficacy and hormone fluctuations across diverse virtual phenotypes based on snapshot baseline compositions.

Open model infrastructure and sharing

To make **MOSAIC** more than a one-off integrated model, we will build it around open, modular software infrastructure for composing, sharing, and coordinating model development across teams. Each model component will have an explicit interface that defines its inputs, outputs, assumptions, units, timescale, and biological scope, allowing microbiome dynamics, metabolic models, agent-based models, and whole-body compartment models to be developed and validated independently while still working together in larger simulations. For general data sharing, we will use common tools for which all our institutions have licenses (e.g., Box). We will use GitHub-based infrastructure for version control, issue tracking, documentation, code review, and coordination across modeling teams, and distribute reusable software components through PyPI where appropriate. Models, parameter sets, simulation workflows, and validation datasets will be packaged with clear provenance and documentation so they can be reused, extended, and compared across teams. We will consider personal health information (PHI) and try to be as streamlined as possible, while also taking university policies into account. This open infrastructure will make **MOSAIC** more reproducible, easier to validate, and adaptable beyond the initial use cases.

Advantages & strengths

As already highlighted above, **MOSAIC** embraces the inter-system complexity and applies the appropriate modeling framework to the associated system rather than forcing the biology to fit into a specific modeling framework. There are apt analogies in many high-functioning engineered systems; an airplane uses optimal materials for its outer shell to maximize aerodynamics and hull integrity and different optimal materials for seat covers to maximize passenger comfort. Similarly, **MOSAIC** capitalizes on the best modeling framework for a given physiological system and then identifies mechanisms of interconnectivity. All done with state-of-the-art model construction and coding practices.

Accounting for sex as a biological variable

Extensive research demonstrates that sex-specific physiological differences significantly influence therapeutic efficacy and toxicity. Building on this, previous work from the Papin lab (41) utilized sex-specific transcriptomic data to reveal distinct metabolic variations, particularly within the pentose phosphate pathway, between male and female genome-scale liver metabolic models. These metabolic differences are hypothesized to modulate individual susceptibility to hepatotoxic pharmaceuticals.

MOSAIC proposes to bridge the gap in existing literature surrounding the mechanistic role of hormone homeostasis in shaping therapeutic response. **MOSAIC** will account for sex as a biological variable through hormone profiling and the incorporation of sex-specific physiology. **The novelty lies in the fact that rather**

than looking at hormone levels in isolation, MOSAIC will assess how hormone dynamics bidirectionally affect metabolism and host-microbiome interactions – all of which integrate to shape an individual's therapeutic response. While current computational frameworks model the gut, liver, and vaginal systems in isolation, our integrated approach captures how sex-specific variables, such as estrogen dynamics, drive disease phenotypes in BV and MASLD. Within this framework, PKPD models will track systemic hormone and drug kinetics to inform constraints for metabolic network models, enabling highly accurate and physiologically relevant metabolic flux predictions. These sex-specific metabolic profiles will define the functional parameters for community and agent-based models. Crucially, this framework establishes a bidirectional relationship where sex-specific differences actively drive the model calibration, whereby population and spatial dynamics continuously re-parameterize the metabolic network and PKPD models. In addition, the PK of drugs and other therapies can be sex-specific, and this will be included where relevant based on clinical pharmacometric studies that measured these differences (42,43).

Assessing the model's validity and credibility

Model credibility will be assessed at multiple levels, including biochemical and physiological consistency, mass balance and feasible flux distributions, and agreement with known sex-specific patterns of metabolism and hormone regulation. Parameter identifiability methods, both structural and practical, will be used to ensure that the data being used to parameterize the model is sufficient for the purpose. Several predictions of the modules will be directly compared to accessible and repeatable clinical measurements, e.g. hormone levels in the blood before and after interventions and plasma drug concentrations over time, and this data will be separated into training data for parameterization and testing data for validation, with no overlap between the two. Uncertainty quantification and uncertainty propagation methods will be used to quantify the robustness of the predictions of the linked models to variability in the input data. Relevance will be assessed by comparing model behavior with independent human datasets associated with MASLD and BV. AI-based sensitivity analyses will be used to identify key drivers, uncertainties, and limitations, informing iterative model refinement.

Scientific impact and clinical relevance

The successful implementation of the proposed work will result in a harmonized computational modeling platform using key modeling approaches. **MOSAIC** will enable identification of potential therapies for metabolic and microbial targets and prediction of adverse drug effects in a sex- and disease-specific manner. Because the approach is also human-focused, results are directly translatable for clinical considerations and therapeutic decisions. Specifically, **MOSAIC** will enable the prediction of the efficacy for therapeutics (drugs or biologics) for BV, which affects millions of women per year, with highly variable rates of responsiveness and recurrence and in which pre- and postmenopausal women have different manifestations. **MOSAIC** will also enable the prediction of therapeutic efficacy of emerging drugs for MASLD which has notable sex-specific differences. Both conditions have strong metabolic and microbiome factors. Both conditions are highly dependent on hormone homeostasis. Importantly, while BV and MASLD are the initial focus, because **MOSAIC** will be developed with best-practices in software design and model construction, the framework will be readily extensible to other diseases. We will focus on predictions of efficacy and adverse effects for the therapeutics metronidazole and *L. crispatus* for BV and resmetirom and GLP-1 RA for MASLD as starting points, with intention to generalize the approach to other interventions as feasible.

Experience and past success of the project team

All Investigators bring extensive experience leading large, complex research teams, with past leadership highlights and the coordination of **MOSAIC** personnel detailed in the Leadership Plan. Each lab will dedicate at least one trainee or staff member to each of the four teams (see Leadership Plan) to ensure cohesive integration and fluid data exchange across projects. Many **MOSAIC** investigators have served as PIs or Directors of large scientific activities (see Biosketches and Leadership Plan). There is a history of collaboration among all members of the **MOSAIC** team.

The team of investigators collectively provides expertise in genome-scale metabolic modeling (Papin), microbiome ecology (Arnold), PKPD modeling (Mac Gabhann), agent-based modeling (Peirce-Cottler), software infrastructure (Agmon), reproductive biology (Derré), liver physiology (Khetani), microbiome-focused clinical research (Young, Vaughan), and experimental validation systems (Papin, Peirce-Cottler, Derré,

Vaughan, Young, Khetani). Together, the team has expertise with each modeling modality represented in **MOSAIC**.

Potential risks and alternative plans

Risks for the proposed modeling efforts include #1) paucity of appropriate experimental data to properly parameterize the models, #2) model predictions that fail validation experiments, and #3) lack of connectivity between modeling frameworks. By design, the **MOSAIC** team is composed of cell biologists and clinicians with the relevant expertise. This expertise ensures that the models are grounded in biological and clinical reality. In addition, these investigators readily generate the data types that the models will use. For risk #1, if existing data is insufficient, we are well positioned to generate the data ourselves (see below). Validating experiments often identify aspects of the model that fail and in fact such failures are often a catalyst to the discovery of new biology. For risk #2, we will approach consistent model prediction failures by limiting the scope of the model (e.g., focus on liver metabolism alone instead of coupling with the role of microbiome metabolism) and pivot the therapeutic focus to drugs that have a more targeted effect. For risk #3, integrated modeling approaches can face challenges with differences in spatial and temporal scales. If we find that the dynamics and parameters that are critical at the interface of the modeling approaches are difficult to identify, we can leverage recently developed AI tools (like graph neural networks (44)) that can extract input/output relationships without explicit accounting for underlying mechanisms. While our preference is to keep many of the underlying models mechanistic because doing so allows us to identify specific molecular targets, we can consider applying graph neural networks or other such approaches to abstract some of those connections to ensure model functionality.

Reuse of existing resources

MOSAIC will serve as an excellent platform to integrate existing multiomics data (metabolomics, transcriptomics, genomics, proteomics) and hormonal and drug dynamic information. The **MOSAIC** investigators are tightly connected with many of the data resources described below as either collaborators with the associated PIs or as contributors to the actual data repositories.

Metagenomics and multiomics of microbial communities

The University of Maryland Baltimore, Human Microbiome Project (UMB-HMP) dataset (BioProject PRJNA208535) contains about 5000 metagenomic profiles of the vaginal microbiome in the absence and presence of menses. The original cohort data contains vaginal swabs from 135 nonpregnant women of reproductive age, collected daily for 10 weeks (45). The dataset has since been expanded to include shotgun sequencing of several thousand vaginal samples. This large dataset will be used to create personalized digital twins of individual vaginal microbiomes and train machine learning models to predict vaginal microbiome compositions over time.

The multi-strain synbiotic vaginal tablet formulation (VS-01) dataset includes shotgun metagenomic profiles of vaginal microbiomes from participants receiving either a probiotic intervention or placebo (46). The CONRAD BV dataset includes 16S metagenomic profiles from participants with bacterial vaginosis (BV) who received metronidazole treatment (47) while the VIBRANT dataset includes shotgun metagenomic profiles from participants with BV who received metronidazole treatment followed by either a probiotic intervention or placebo (48). Metatranscriptomics of the vaginal microbiome in BV patients treated with metronidazole is available under PRJEB21446 and will be used to further profile models of the vaginal microbiome (49). Metabolomics of the vaginal microbiome in women with and without BV will be used to parameterize metabolic conditions in the vaginal environment (50). Together, these datasets provide longitudinal profiles of vaginal microbiome responses to therapeutic interventions and will be used to train and evaluate machine learning models for predicting microbiome dynamics following treatment.

The Human Microbiome Project (HMP) dataset will also be used to generate *in silico* clinical cohorts representative of the healthy gut microbiome. The HMP1-II project (PRJNA48479) contains 249 metagenomic profiles from healthy individuals and provides a cross-sectional characterization of gut microbial composition (51). The Broad Institute–OpenBiome Microbiome Library (BIO-ML) (PRJNA544527) includes longitudinal stool metagenomic data from 80 healthy participants (52). Metabolomics were also run on samples collected at various time points over the 18 months. The BIO-ML dataset will complement the HMP cohort by providing longitudinal data to capture temporal variability in the healthy gut microbiome.

A multiomics dataset, including 16S metagenomics, transcriptomics, and metabolomics of stool samples from NAFLD-diagnosed and control participants will be used to model the gut microbiome in

individuals with and without MASLD (53). The dataset includes samples from men and postmenopausal women. A larger dataset (PRJNA629344) containing 16S and shotgun metagenomics of MASLD and control participants can be used to increase the power of metagenomic analysis if needed (54).

Host cell transcriptomics

The BARICO dataset (PRJNA1051582) provides bulk RNA-seq transcriptomic profiles from liver biopsies of individuals diagnosed with metabolic dysfunction-associated steatotic liver disease (MASLD) (17). An additional study (PRJNA1154708) includes liver transcriptomic profiles and plasma proteomic profiles from participants with MASLD and healthy controls (55). Both datasets include male and female representation. Together, these datasets provide complementary host molecular profiles that will be used to investigate host alterations associated with MASLD. RNAseq transcriptomic data (PRJNA731542) from ectocervical biopsies of individuals with BV that have been given metronidazole will be used to profile cervicovaginal cell models (56). A subset of the HuBMAP collection (HBM692.JRZB.356) will be used to profile gut epithelial cell models of healthy donors. This dataset includes scRNA-seq transcriptomics of 8 different collection sites in the gut (57). There are currently no published datasets of gut epithelial transcriptomics in individuals with BV or MASLD.

Hormone profiling and other data sets

The mcPHASES dataset will be used to model levels of estrogen and progesterone during the menstrual cycle (58). mcPHASES includes daily urinalysis of hormone metabolites across a three-month observation period. All data is publicly available on Physionet. The MESA AC 291 dataset (BioLINCC HLB00640825a) will be used to analyze levels of bioavailable testosterone, estradiol, dehydroepiandrosterone and sex hormone binding globulin associated with NAFLD (59). This subset of MESA includes data from postmenopausal women and men. We will use published pharmacokinetic (PK) profiles of resmetirom (42) and metronidazole (43) to inform model parameterization and drug exposure dynamics. We will use AERSMine (60), an application which makes the Food and Drug Administration's Adverse Event Reporting System (AERS) more accessible, as we have done in the past (41). AERSMine can be used to analyze trends in side effect reports by specific drug, adverse event, sex, and age.

Primary data collection

All primary data collection will be model-driven, restricted to parameterization gaps or independent validation of model predictions. We plan to use the human-centered *in vitro* models outlined below to generate these data where the publicly available datasets do not provide all of the data necessary for model input or where specific validations will be performed (e.g., drug testing, hormone levels). Experiments with such i*n vitro* systems can be integral to identifying areas in the models where further curation may be required to improve predictions. Regarding human subject research, an observational retrospective study to collect vaginal swabs is already IRB approved and active at UVA. Through this study, we can collect clinical data through electronic medical records and match vaginal swab samples. The current study is limited by being a cross-sectional study and there is no patient contact. If there is a need to develop a prospective longitudinal study to gather samples that can better inform the models (over *in vitro* approaches), then we will develop and go through the approval process. A prospective longitudinal study has the potential to provide valuable data as to the personalization of the **MOSAIC** platform.

Rigorous model validation will require independent experimental testing. A key characteristic of the **MOSAIC** modeling framework is to embed a person with each type of modeling expertise (metabolic network models, community models, agent-based models, PKPD models, multi-scale models) on each application team (BV, MASLD) to ensure there is actual crosstalk between the modeling frameworks. In addition, each application team (BV, MASLD) has a cell biologist and a clinician with the respective expertise to ensure models are constructed with awareness of the underlying biology and relevant drug targets from the very beginning of model construction.

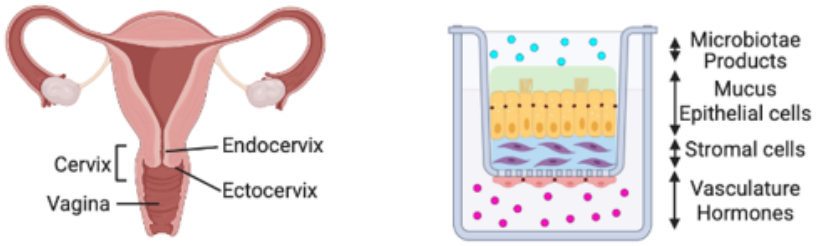


Figure 3. An *in vitro* model of the female reproductive tract consists of epithelial and stromal cells interfacing microbial and hormonal components. Derré has developed this cell culture system with remarkable consistency with human physiology. This system can serve as an important testbed for model parameterization and validation as necessary.

Reproductive Tract Model (Figure 3): To model the effects of hormonal regulation within the human female reproductive tract, we established a stepwise multicellular Transwell platform composed of human vaginal or endocervical epithelial cells (cell lines or primary cells) cultured on the apical surface of a collagen matrix containing SHT290 fibroblasts. The system can be further expanded by incorporating endothelial cells on the basolateral side of the membrane to increase physiological complexity. Following establishment of the model, hormones can be administered to the basolateral compartment to mimic systemic exposure, while microbiota-derived products or live bacteria can be applied to the apical epithelial surface to model host–microbe interactions. This modular platform allows the addition or removal of specific cell types as needed and is readily compatible with a range of downstream analyses (61,62). Potential therapeutics to be examined in the *in vitro* reproductive tract model include the standard of care for BV (metronidazole) and a biologic/probiotic (*Lactobacillus crispatus*) which have been used in clinical trials (63,64).

Human 3D liver microtissues (Figure 4)*:* The Khetani lab utilizes droplet microfluidics to fabricate extracellular matrix protein-based microtissues containing primary human hepatocytes (PHHs) (65–67) with hepatic stellate cells (HSCs), or liver sinusoidal endothelial cells (LSECs). Microtissues conserve the use of limited primary human liver cells, and exhibit 3D cell-cell and cell-ECM interactions, unlike 2D monolayers used in existing liver-on-a-chip (microfluidic) devices (68–73). Using various combinations, noninflamed and inflamed (MASLD) states of the microtissues can be generated as shown by high levels of inflammatory cytokines and other inflammatory markers that are consistent with clinical studies in MASH patients due to inflammation (74,75).

Gut-liver model (Figure 4)*:* Similar to the reproductive tract model, a transwell configuration using primary human intestinal stem cells (membrane layer to establish polarity) and fibroblasts and PHHs (bottom of well) can be used to examine how the intestinal microbiome interacts with the liver and gut cells using TEER (transepithelial resistance) and CYP3A4 levels to monitor intestine and liver cell activity (76–80). Two FDA approved drugs of interest to test *in vitro* include resmetirom and GLP-1 receptor agonists. Resmetirom is a thyroid hormone receptor beta agonist and is used in adults with noncirrhotic MASH (81,82). Use of the Glucagon-like peptide-1 receptor agonist (GLP-1 RA, semaglutide) in patients with MASLD and MASH exhibited sex-dependent and independent differences in responses (8,83). The hepatic benefit is largely mediated through improved insulin sensitivity, reduced delivery of lipotoxic substrates, and the gut-liver-microbiome axis (84). Overall, MASLD is sexually dimorphic, with the estrogen, androgen, and thyroid axes each modulating hepatic lipogenesis, fatty acid oxidation, and fibrogenesis likely affecting gut epithelial cells and the microbiota.

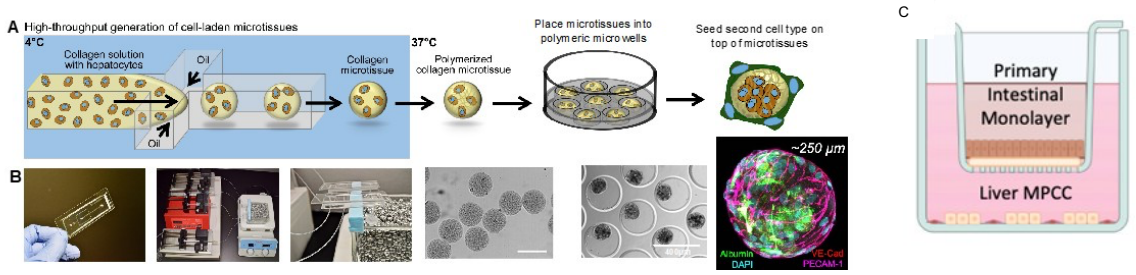


Figure 4. (A) To represent liver tissue function, a mixture of ECM and cells is perfused into a microfluidic droplet generator at specific experimental conditions to enable microstructures consistent with liver tissue physiology (**A**). Polydimethylsiloxane (PDMS) microfluidic devices with a single emulsion droplet generator to facilitate the creation of the liver tissue models **(B).** An intestinal epithelial monolayer with cocultured micropatterned cells (MPCC) in a Transwell format to simulate the gut influence on liver function (**C**).

Conclusion

MOSAIC is a potential paradigm-shifting strategy for modeling human physiology. Rather than developing a single modeling framework to represent varied biology, **MOSAIC** embraces the reality of interconnected systems that underlie health and disease. Hormone homeostasis is embedded in every facet of **MOSAIC.** For example, transcriptional profiles as a function of estrogen levels will be integrated into the metabolic network models, PKPD models will explicitly account for estrogen dynamics, microbial community models will be derived from data that captures dynamics of menses, and models will be constructed as a function of hormonal status (e.g., pre/post menopause). The team of investigators possess the ideal combination of modeling expertise and associated biological and clinical expertise. **MOSAIC** will be well-positioned to predict sex-specific therapies for BV and MASLD and lay the foundation for application to other diseases at the interface of human metabolism and the microbiome.

Project Team. Included are all team leaders and other trainees and staff already identified. Further personnel will be appointed to additional roles as outlined in the Budget Justification.

Last Name	First Name	eRA Commons ID	Institutional affiliation	Current title	Email address	Percent effort	Role in the project
Papin	**Jason**	papinnih	University of Virginia	Harrison Distinguished Professor of Biomed. Eng.	papin@virginia.edu	42%	**Overall PI** **Metabolic Network Modeling Lead**
Arnold	**Kelly**	kbarnold	University of Michigan	Associate Professor of Biomed. Eng.	kbarnold@umich.edu	25%	**Dynamic Modeling Lead**
Peirce-Cottler	**Shayn**	smp6pnih	University of Virginia	Chair and Harrison Distinguished Professor of Biomed. Eng.	shayn@virginia.edu	25%	**ABM/Integration Modeling Lead**
Mac Gabhann	**Feilim**	fm9eNIH	Johns Hopkins University	Associate Professor of Biomed. Eng.	feilim@jhu.edu	25%	**Compartment/ PK-PD Model Lead**
Agmon	**Eran**	eranagmon	University of Connecticut Health	Assistant Professor of Molecular Biology and Biophysics	agmon@uchc.edu	25%	**Software Development Lead**
Khetani	**Salman**	skhetanipi	University of Illinois, Chicago	Robert Uyetani Collegiate Professor of Biomed. Eng.	skhetani@uic.edu	10%	**GI/Liver Cell Biology Expert**
Derre	**Isabelle**	iderre	University of Virginia	Associate Professor of Microbiology, Immunology and Cancer Biology	id8m@virginia.edu	10%	**Vaginal Cell Biology Expert**

Vaughan	**Monique**	mhvaughan	University of Virginia	Assistant Professor of Obstetrics & Gynecology	mv4w@uvahealth.org	10%	**BV Clinical, Microbiome Expert**
Young	**Vince**	youngvi	University of Michigan	William Henry Fitzbutler Collegiate Professor	youngvi@umich.edu	10%	**GI Clinical, Microbiome Expert**
Kolling	**Glynis**	glk3anih	University of Virginia	Research Associate Professor of Biomed. Eng.	glk3a@virginia.edu	50%	**Program Manager**
Felton	**Samantha**	n/a	University of Virginia	Postdoctoral Fellow	smf2qp@virginia.edu	100%	**Integrative Modeler**
Tantry	**Avani**	n/a	University of Virginia	Graduate Student	TBD	100%	**VMB Metabolic Modeler**
Wettstone	**Erin**	n/a	University of Virginia	Graduate Student	egw6vx@virginia.edu	100%	**GI Microbiome Metabolic Modeler**
Moore	**Connor**	n/a	University of Virginia	Graduate Student	cjm4yy@virginia.edu	100%	**GI Epithelial Metabolic Modeler**
Bonakdar	**Slna**	n/a	University of Michigan	Graduate Student	bonakdar@umich.edu	100%	**VMB Community Modeler**
Rahman	**Tasnif**	n/a	University of Connecticut Health	Postdoctoral Fellow	trahman@uchc.edu	100%	**Multiscale Software Developer**
Palacios-Gomez	**Christian**	n/a	University of Virginia	Graduate Student	cp4qxz@virginia.edu	100%	**GI ABM Modeler**
Pandey	**Regeant**	n/a	University of Illinois, Chicago	Research Associate	rpanda3@uic.edu	100%	**GI/Liver Cell Biology Experimenter**

Subrecipients

Organization	Project role
University of Michigan	dynamic modeling, microbiome analysis
Johns Hopkins University	compartment modeling, PK/PD modeling, model integration
University of Connecticut	software development
University of Illinois, Chicago	GI/liver cell experimentation

OMB No. 0925-0001 and 0925-0002 (Rev. 10/2021 Approved Through 01/31/2026)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
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NAME: Agmon, Eran

eRA COMMONS USER NAME (credential, e.g., agency login): eranagmon

POSITION TITLE:Assistant Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of California San Diego, San Diego, CA, USA	BACHELOR OF SCIENCE	03/2009	Cognitive Science, Neuroscience
Portland State University, Portland, OR, USA	MASTER OF SCIENCE	06/2011	Systems Science
Indiana University, Bloomington, IN, USA	DOCTOR OF PHILOSOPHY	08/2016	Informatics and Computing, Cognitive Science
Columbia University, New York, NY, USA	Postdoctoral Fellow	10/2017	Chemical Biology, Computational Biology
Stanford University, Stanford, CA, USA	Postdoctoral Fellow	07/2022	Bioengineering

A. Personal Statement

My overarching research goal is to pioneer comprehensive, multi-scale simulations of cellular systems, from molecular biology to tissues and organ scale. Over the first decade of my career, I laid the groundwork, working in various domains of computational biology including molecular dynamics of lipid membranes, constraint-based models of metabolic networks, stochastic models of genetic networks, cell shape with a cellular Potts model, adaptive signal processing with spiking neural networks, and a recurrent neural network that controls agent navigation. These experiences enriched my understanding of biological phenomena and their different modeling approaches. As a postdoc, I contributed to the next generation E. coli whole-cell model, harnessing 19,000 experimentally-derived parameters in a system with over 10,000 equations and 16,000 molecular species [Macklin, 2020]. To streamline the project's framework, I developed the software Vivarium [Agmon, 2022], a model integration tool that connects simulation methods and models, enabling compositional multi-scale modeling across all scales of biology, which remade the E. coli model and allowed it to scale up to colony simulations and response to environmental shifts in antibiotics [Skalnik, 2023]. We recently applied Vivarium to tumor-scale models with multiple cell types and locations, retrieving initial conditions from CODEX spatial imaging datasets [Hickey, 2024]. I have become a leader in integrative multiscale modeling – an approach that combines diverse simulators with different models of subsystems and a variety of multi-omics datasets. My reasoning is that instead of using a single class of mathematical representation to capture a narrow range of behavior, we should build detailed models of many interacting mechanisms – each with the most appropriate, meaningful representation. To simplify how we build these models, I led the development of software tools for combining multi-source and multi-level data with models of diverse mechanisms to build predictive models of biological systems as integrated wholes.

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

2022 - present Assistant Professor, University of Connecticut Health, Farmington, CT, USA
2017 - 2022 Postdoctoral Fellow, Stanford University, Stanford, CA, USA
2016 - 2017 Visiting Scholar, Institute for Advanced Study, Princeton, NJ, USA

2016 - 2017 Postdoctoral Research Scientist, Columbia University, New York City, NY, USA
2011 - 2016 Graduate Student, Indiana University, Bloomington, IN, USA

Honors
2020 - 2022 F32 Postdoctoral Fellowship, NIH
2016 Cognitive Science Research Fellowship, Indiana University
2016 Outstanding Dissertation Award, Indiana University Cognitive Science
2015 Best paper award, European Conference on Artificial Life
2015 1st place poster, IGERT Research Showcase
2014 Outstanding Teaching Award, Indiana University Cognitive Science
2011 - 2016 IGERT Fellowship in the Dynamics of Brain-Body-Environment Systems, NSF

C. Contributions to Science

1. Software for model integration. I led the development of a novel, open-source model integration tool called Vivarium. The aim was to make it easier for scientists to define any imaginable mechanistic model, combine it with existing models, and execute them together as an integrated simulation. It provides an interface that makes individual simulation tools into modules that can be wired together, parallelized across multiple CPUs, and simulated across many spatial and temporal scales. Vivarium establishes a modular design methodology that supports flexible model development – users can add, remove, swap, and reconfigure the previous modules. Building comprehensive models of cells will require many contributors to continue building upon and revisiting prior knowledge. To advance this vision, I started the Vivarium Collective – a registry of reusable Vivarium modules that can be reconfigured to prototype new models and iterate on model design.

2. Hybrid mechanistic modeling for whole-cell and multicellular modeling. I led a team to implement the complete E. coli whole-cell model with Vivarium, integrating over a dozen independent sub-models, thus creating a valuable resource for the broader modeling community. Projects under my leadership have showcased the gradual layering of complexity, including a model of chemotactic E. coli that incorporated a multitude of computational techniques, enabling the simulation of navigation in nutrient-rich environments and adjusting growth rates. We further connected a genome-scale metabolic model of E. coli with a stochastic model of transporter expression and glucose/lactose uptake, leading to a metabolic shift unattainable by any single model. With the Allen Center for Systems Modeling of Infection, I linked the whole-cell model of E. coli with agent-based modeling to simulate colony growth, antibiotic resistance, and hundreds of E. coli cells in a spatial environment. We recently applied Vivarium to tumor-scale models with multiple cell types and locations, retrieving initial conditions from CODEX spatial imaging datasets.

3. Vision for next-generation predictive biology. I have advocated for advanced infrastructure to support large-scale modeling in biology. In these publications, I provide a vision for what kind of infrastructure will be required, drawing inspiration from meteorology and from machine learning models.

4. Modeling the Lipid Basis of Ferroptosis. Ferroptosis is a form of cell death different from apoptosis, mostly examined through methods of chemical biology by exposing cells to different compounds and observing their trajectory to death. There was a hypothesis that ferroptosis uses reactive oxygen species to peroxidize unsaturated phospholipids, and this leads to cell death but the mechanism by which this leads to death was not known. I worked on a molecular dynamics model of membranes to investigate what properties change when lipids become peroxidized, and whether membrane changes are sufficient to explain cell death. I adopted an open source coarse-grain molecular dynamics software called MARTINI, and developed a peroxide subunit that could be used to modify phospholipids. The resulting model demonstrated how membrane properties change when they undergo peroxidation in a way that promotes membrane blebbing and pore formation (Agmon & Stockwell 2017). These predictions were later verified by a team of biochemists (Agmon et al. 2018).

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Kelly B. Arnold

eRA COMMONS USER NAME (credential, e.g., agency login): kbarnold

POSITION TITLE: Associate Professor of Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
Rice University, Houstin, TX	B.S.	05/2004	Bioengineering
University of Virginia	Ph.D.	06/2010	Biomedical Engineering
Massachusetts Institute of Technology	Postdoctoral Fellow	06/2015	Biological Engineering

A. Personal Statement

We are excited to contribute to the work described in the proposal, where we will update our existing mechanistic microbiome models to automate production of digital twins based on an individual's specific microbiome composition. We will integrate the models with metabolic models in the Papin lab to identify metabolic events that underly key microbial interactions related to hormone homeostasis and drug efficacy. We will also collaborate with other project Co-Is to adapt the approach for use in any tissue/microbiome community.

B. Positions, Scientific Appointments, and Honors

2022 - present Associate Professor, Department of Biomedical Engineering, University of Michigan
2015 - 2022 Assistant Professor, Department of Biomedical Engineering, University of Michigan
2010 – 2015 Postdoctoral Fellow, Department of Biomedical Engineering, Massachusetts Institute of Technology

C. Contributions to Science

1. Dynamic models of the vaginal microbiome for insight into therapeutic efficacy and community resilience. We have developed some of the first ordinary differential equation (ODE) models to identify new systems-level mechanisms that link community features of the vaginal microbiome to variability in disease susceptibility and therapeutic efficacy. These models have allowed us to reveal previously unappreciated microbial features that vary among women and account for differences in responses to therapeutic treatments and to hormonal changes from menses. These models also provide a new framework for hypothesis-testing related to future rational design of vaginal antibiotic and probiotic therapy.

a. Lee CY*, Cheu RK*, Lemke MM, Gustin AT, France MT, Hampel B, Thurman AR, Doncel GF, Ravel J, Klatt NR[#], **Arnold KB[#]**. Quantitative modeling predicts mechanistic links between pre-treatment microbiome composition and metronidazole efficacy in bacterial vaginosis. Nat Commun. 2020 Dec 1;11(1):6147. doi: 10.1038/s41467-020-19880-w. PMID: 33262350; PMCID: PMC7708644.

b. Lee CY, Diegel J, France MT, Ravel J, **Arnold KB**. Evaluation of vaginal microbiome equilibrium states identifies microbial parameters linked to resilience after menses and antibiotic therapy. PLoS Comput Biol. 2023 Aug 11;19(8):e1011295. doi: 10.1371/journal.pcbi.1011295. PMID: 37566641; PMCID: PMC10446192.

c. Lee CY, Bonakdar S, **Arnold KB**. An in silico framework for the rational design of vaginal probiotic therapy. bioRxiv 2024.04.10.588978; doi: https://doi.org/10.1101/2024.04.10.588978.

d. Lee CY, Dillard LR, **Papin JA**, **Arnold KB**. New perspectives into the vaginal microbiome with systems biology. Trends Microbiol. 2023 Apr;31(4):356-368. doi: 10.1016/j.tim.2022.09.011. Epub 2022 Oct 19. PMID: 36272885. *Selected as cover feature.*

e. Cheu RK, Gustin AT, Lee C, Schifanella L, Miller CJ, Ha A, Kim C, Rodriguez VJ, Fischl M, Burgener AD, **Arnold KB**, Alcaide ML, Klatt NR. Impact of vaginal microbiome communities on HIV antiretroviral-based pre-exposure prophylaxis (PrEP) drug metabolism. PLoS Pathog. 2020 Dec 3;16(12):e1009024. doi: 10.1371/journal.ppat.1009024. PMID: 33270801; PMCID: PMC7714160.

2. Data-driven models to infer mucosal proteomic networks involved in disease progression and therapeutic efficacy. My team has developed data-driven models to reveal immune networks involved in disease progression in mucosal tissues. These models integrate high-throughput proteomic, transcriptomic, and/or microbiome data from clinical samples to reveal new networks driving disease susceptibility and therapeutic efficacy

f. DiLillo KM, Ruvuna L, Pratte KA, Serban KA, Labaki WW, Han MK, Freeman CM, Bowler RP, Curtis JL, **Arnold KB**. Validation of Systemic Complement Signatures in the Progression of Chronic Obstructive Pulmonary Disease. Am J Respir Crit Care Med. 2024 Nov 15;210(10):1269-1272. doi: 10.1164/rccm.202311-2059LE. PMID: 39311978; PMCID: PMC11568447.

g. DiLillo KM, Norman KC, Freeman CM, Christenson SA, Alexis NE, Anderson WH, Barjaktarevic IZ, Barr RG, Comellas AP, Bleecker ER, Boucher RC, Couper DJ, Criner GJ, Doerschuk CM, Wells JM, Han MK, Hoffman EA, Hansel NN, Hastie AT, Kaner RJ, Krishnan JA, Labaki WW, Martinez FJ, Meyers DA, O'Neal WK, Ortega VE, Paine R 3rd, Peters SP, Woodruff PG, Cooper CB, Bowler RP, Curtis JL[#], **Arnold KB[#]**; SPIROMICS investigators. A blood and bronchoalveolar lavage protein signature of rapid FEV_1 decline in smoking-associated COPD. Sci Rep. 2023 May 22;13(1):8228. doi: 10.1038/s41598-023-32216-0. PMID: 37217548; PMCID: PMC10203309.

h. D O'Dwyer DN, Ashley SL, Gurczynski SJ, Xia M, Wilke C, Falkowski NR, Norman KC, **Arnold KB**, Huffnagle GB, Salisbury ML, Han MK, Flaherty KR, White ES, Martinez FJ, Erb-Downward JR, Murray S, Moore BB, Dickson RP. Lung Microbiota Contribute to Pulmonary Inflammation and Disease Progression in Pulmonary Fibrosis. Am J Respir Crit Care Med. 2019 May 1;199(9):1127-1138. doi: 10.1164/rccm.201809-1650OC. PMID: 30789747; PMCID: PMC6515865.

3. Computational models to unravel heterogeneity in antibody responses after vaccination and infection. We have used data-driven and ODE models to uncover new mechanisms that account for variability in disease severity and the immune response to vaccination**.**

i. Shoffner-Beck SK, Theisen RM, Wong KE, Rerks-Ngarm S, Pitisuttithum P, Nitayaphan S, Kent S, Chung AW, **Arnold KB**. Understanding mechanistic relationships between IgG titers and Fc effector functions: a computational framework to assess polyfunctionality. Front Immunol. 2025 Sep 16;16:1578500. doi: 10.3389/fimmu.2025.1578500. PMID: 41035650; PMCID: PMC12479434.

j. Lemke MM, McLean MR, Lee CY, Lopez E, Bozich ER, Rerks-Ngarm S, Pitisuttithum P, Nitayaphan S, Kratochvil S, Wines BD, Hogarth PM, Kent SJ, Chung AW[#], **Arnold KB[#]**. A systems approach to elucidate personalized mechanistic complexities of antibody-Fc receptor activation post-vaccination. Cell Rep Med. 2021 Sep 1;2(9):100386. doi: 10.1016/j.xcrm.2021.100386. PMID: 34622227; PMCID: PMC8484512.

k. Lemke MM, Theisen RM, Bozich ER, McLean MR, Lee CY, Lopez E, Rerks-Ngarm S, Pitisuttithum P, Nitayaphan S, Kratochvil S, Wines BD, Hogarth PM, Kent SJ, Chung AW, **Arnold KB**. A Quantitative Approach to Unravel the Role of Host Genetics in IgG-FcγR Complex Formation After Vaccination. Front Immunol. 2022 Feb 22;13:820148. doi: 10.3389/fimmu.2022.820148. PMID: 35273603; PMCID: PMC8902241.

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BIOGRAPHICAL SKETCH

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NAME: Isabelle Derré

eRA COMMONS USER NAME (credential, e.g., agency login): iderre

POSITION TITLE: Associate Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University Paris VII, France	B.S	06/1996	Biology & Genetics
Institute Pasteur, Paris, France	Diploma	08/1996	Microbiology
University Paris VII, France	M.S	08/1997	Microbiology
University Paris VII, France	Ph.D.	09/2000	Microbiology
Tufts University, USA	Postdoctoral	03/2004	Host-Pathogen Interaction
Pasteur Institute, France	Postdoctoral	12/2004	*Chlamydia* Pathogenesis

A. Personal Statement

I am a tenured Associate Professor in the Department of Microbiology, Immunology, and Cancer Biology at the University of Virginia. My research program focuses on the sexually transmitted intracellular pathogen *Chlamydia trachomatis*, with particular expertise in host-pathogen interactions, microbial genetics, and infection biology. My research seeks to understand how *C. trachomatis* interacts with its host to establish infection and cause disease. Through sustained NIH-supported work spanning microbial genetics, cell biology, animal models, organoid systems, and biomimetic human tissue models, I have developed expertise in studying host-pathogen interactions in increasingly complex and physiologically relevant contexts. These experiences, combined with a strong interest in applying novel computational and analytical approaches to biologically meaningful questions, provide a strong foundation for my contributions to the proposed MOSAIC project.

B. Positions, Scientific Appointments, and Honors

2023-present Tenured Associate Professor, MIC Department, University of Virginia, Charlottesville VA
2020-2023 Associate Professor, MIC Department, University of Virginia, Charlottesville VA
2015-2020 Assistant Professor, MIC Department, University of Virginia, Charlottesville VA
2014 -2015 Research Scientist, Agaisse Laboratory, Yale University, New-Haven CT
2005 -2014 Associate Research Scientist, Agaisse Laboratory, Yale University, New-Haven CT
2004-2005 Post-doctoral Fellow, Dautry-Varsat Laboratory, Pasteur Institute, Paris, France
2001-2004 Post-doctoral Fellow, Isberg Laboratory, Tufts University School of Medicine, Boston, MA
1997-2000 Graduate Student, Unité de Biochimie Microbienne, Pasteur Institute, Paris, France
1997 Masters Student, Unité de Biochimie Microbienne, Pasteur Institute, Paris, France

C. Contributions to Science

1- Development of organoid and microphysiologic models to study *Chlamydia* infection
To better model *Chlamydia* pathogenesis in the female reproductive tract, we have developed murine endometrial organoids (NIAID R03 AI146649; PMID: 32923409) and a human cervical microphysiologic model (NIAID U19 AI158930; PMID: 38103413, 41931629). These models recapitulate tissue architecture and host-pathogen interactions, providing a more physiologically relevant context than traditional cell culture. Ongoing work focuses on using these systems to study bacterial infection, cellular responses, and the factors, such as sex hormones, that influence *Chlamydia* colonization and replication. Collectively, these efforts provide innovative tools to advance mechanistic studies of *Chlamydia* infection and to inform future translational research.

2- Characterization of direct membrane contact sites (MCS) between the Chlamydia inclusion membrane and the endoplasmic reticulum (ER).
Chlamydia are obligate intracellular pathogens that rely on the host cell for replication. Using high-throughput genome-wide siRNA screens, we identified host factors that affect *Chlamydia* infection. We discovered that, in addition to manipulating vesicular trafficking, *Chlamydia trachomatis* hijacks non-vesicular lipid transfer at specialized ER–inclusion membrane contact sites (MCS), regions of close apposition (10–50 nm) between the ER and inclusion membrane (NIAID R01 AI101441; PMID:21731489). We identified protein complexes composed of *Chlamydia* inclusion membrane proteins (Inc proteins) and host factors and characterized their roles in ER tethering (NIAID R01 AI101441; NIAID R21 AI141841; PMID: 29078338, 35838228), lipid trafficking (NIAID R01 AI101441; NIAID R01 AI189455; PMID:21731489), and inclusion integrity (NIAID R01 AI162758; PMID: 36853051). Our ongoing work defines how these bacterial-host protein complexes establish the intracellular niche and mediate *Chlamydia* pathogenesis in vivo.

3- Development of genetic tools to study the *Chlamydia* developmental cycle.
Genetic intractability has long limited functional studies of *Chlamydia* virulence factors. We developed cloning vectors for transgene expression in *C. trachomatis*, generating strains expressing fluorescent proteins for live-cell imaging of the developmental cycle (NIAID R01 AI101441; PMID: 31170215). To address protein toxicity, we created a conditional gene expression system to regulate expression of tagged virulence factors, allowing generation of conditional mutants to study essential genes (NIAID R01 AI162758; NIAID R01 AI189455; PMID: 36084160). We engineered similar tools for *C. muridarum*, enabling in vivo studies. In addition, we developed a toxin/antitoxin-based counterselection marker for *C. trachomatis*, facilitating allelic exchange and expanding the chlamydial genetic toolkit (NIAID R01 AI162758; PMID: 41251379). These advances expanded the *Chlamydia* genetic toolkit and enabled functional analysis of virulence determinants.

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BIOGRAPHICAL SKETCH

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NAME: Khetani, Salman R.

eRA COMMONS USER NAME (credential, e.g., agency login): skhetanipi

POSITION TITLE: Robert Uyetani Collegiate Professor of Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE	Completion Date MM/YYYY	FIELD OF STUDY
Marquette University, Milwaukee, Wisconsin	B.S	05/2000	Biomedical Engineering
Marquette University, Milwaukee, Wisconsin	B.S	08/2000	Electrical and Computer Engineering
University of California, San Diego, California	M.S	08/2002	Bioengineering
University of California, San Diego, California	Ph.D	03/2006	Bioengineering
Massachusetts Institute of Technology	PostDoc	09/2008	Health Sciences and Technology

A. Personal Statement

My expertise is in developing long-lasting in vitro cultures of human cells (liver, cardiac, intestine, and placenta) for various applications, using semiconductor-driven microfabrication tools. After my postdoctoral studies at MIT in the lab of Dr. Sangeeta Bhatia, I co-founded and led research at a startup (Hepregen Corporation, now part of BioIVT) to further develop micropatterned liver co-cultures for drug-screening applications. In 2011, I returned to academia as an assistant professor at Colorado State University, where my lab continued to develop more mature/advanced liver models and investigated the mechanisms of liver diseases in vitro, including hepatitis C and B viral infections, as well as alcoholic and metabolic dysfunction-associated fatty liver diseases. We have also gained experience in the culture of liver nonparenchymal cells, including hepatic stellate, Kupffer, and liver sinusoidal endothelial cells. We are using this experience to develop next-generation liver platforms using microfluidics, co-culture with non-parenchymal liver cells, and biomaterial approaches. Effective August 2015, I moved to UIC to join the Department of Biomedical Engineering as a faculty member (associate and now full professor), where my lab continues to innovate in liver tissue engineering and has expanded our expertise to cardiac, intestinal, and placental tissue engineering. My publications so far have been highly cited (7979 citations and an h-index of 40 as of May 2026), and my lab has been funded by NIH and NSF since 2008. My lab is well-suited for this proposal, given our extensive experience with culturing primary human liver cells in various culture formats and using them for studying MASLD and other liver diseases.

B. Positions, Scientific Appointments, and Honors

Positions

2024-Present	Robert Uyetani Professor of Biomedical Engineering, University of Illinois Chicago
2026	Acting Associate Dean for Graduate Studies, University of Illinois Chicago
2025-2026	Associate Head of Biomedical Engineering, University of Illinois Chicago
2022-2024	Professor of Biomedical Engineering, University of Illinois Chicago
2021-2025	Director of Graduate Studies, Biomedical Engineering, University of Illinois Chicago
2018-2022	Tenured Associate Professor of Biomedical Engineering, University of Illinois Chicago
2015-2018	Untenured Associate Professor of Biomedical Engineering, University of Illinois Chicago
2011-2016	Member, Scientific Advisory Board, Hepregen Corporation (now Ascendance Biotechnology), Medford, MA
2011-2015	Assistant Professor of Mechanical Engineering, Colorado State University, Fort Collins, CO
2008-2011	Co-founder and Director of Research, Hepregen Corporation, Medford, MA

Honors

2026-	Fellow, American Institute for Medical and Biological Engineering
2026	Distinguished Researcher, Natural Sciences and Engineering category (awarded to 1 faculty member each year), University of Illinois Chicago
2025-2028	University Scholar (awarded to 6 faculty members each year), University of Illinois Chicago
2023	Departmental Faculty of the Year Award, University of Illinois Chicago
2023	Graduate Student Mentoring Award, Graduate College, University of Illinois Chicago
2023	College of Engineering Teaching Award, University of Illinois Chicago
2022	College of Engineering Teaching Award, University of Illinois Chicago
2018	Departmental Rising Star Award, University of Illinois College of Medicine
2018	Honoring Our Professors' Excellence (HOPE) Award, University of Illinois Chicago
2017	College of Engineering Teaching Award, University of Illinois Chicago
2015	School of Biomedical Engineering Teaching Excellence Award, Colorado State Univ.
2014-2019	National Science Foundation CAREER award

C. Contributions to Science

Experimental Modeling of Metabolically Dysfunction-Associated Steatotic Liver Disease (MASLD): My lab develops engineered human liver models to study metabolic dysfunction–associated steatotic liver disease (MASLD) and its progression to steatohepatitis, fibrosis, cirrhosis, and hepatocellular carcinoma. Because species-specific differences in drug metabolism and disease pathways limit the translatability of animal studies, we engineer multicellular platforms that maintain primary human hepatocytes (PHHs) in a well-differentiated, functionally stable state for weeks to months. Using these systems, we demonstrated that primary human hepatic stellate cells (HSCs) differentiated into fibrotic myofibroblasts modulate PHH phenotype in a dose-dependent manner: increasing HSC numbers downregulated cytochrome-P450 and transporter activities and induced steatosis, while elevating pro-inflammatory IL-6 and C-reactive protein secretion (PMID: 28702667). We established that HSC paracrine signaling, partly via IL-6, mediates these effects, and that clinically relevant NADPH oxidase inhibition and farnesoid X receptor activation alleviate hepatic dysfunction, recapitulating steatohepatitis and early fibrosis-like features with utility for therapeutic discovery. We have also used these models to investigate the metabolic drivers of MASLD, showing that PHHs exposed to hyperglycemic conditions accumulate hepatic lipids and develop selective insulin resistance, linking the pathophysiology of type 2 diabetes to fatty liver disease in vitro (PMID: 27312339); Most recently, we developed high-throughput, droplet-generated 3D liver microtissues that sustain stable hepatic function for six-plus weeks and respond to drug-mediated CYP induction and hepatotoxicity in a clinically relevant manner (PMID: 32290899). We are now incorporating hepatic stellate cells, hepatocytes, and liver sinusoidal endothelial cells into these microtissues to build multicellular MASLD models that recapitulate key features of disease progression and accelerate therapeutic discovery.

Experimental Modeling of Drug Metabolism and Toxicity: Beyond disease modeling, my laboratory applies engineered human liver models to predict drug metabolism and toxicity across the discovery pipeline. Accurate toxicity assessment is integral to identifying compounds with optimal efficacy and minimal side effects, since a drug's therapeutic value depends not only on its activity against disease but on a favorable safety and metabolic profile. The extended longevity of these models enables continuous drug incubations far longer than the 4–6 hours possible with conventional hepatocyte suspensions, allowing accurate evaluation of slowly metabolized compounds. Using these systems, we predicted in vivo hepatic clearance of low-to-intermediate clearance drugs within two- to three-fold of observed human values, with the highest accuracy for low-clearance compounds that are poorly characterized by suspension assays; our model systems generated the major human Phase I and Phase II metabolites, including those arising via sequential reactions, at a performance exceeding hepatocyte suspensions, S-9 fractions, and liver microsomes (PMID: 30552098). These models also support sensitive prediction of drug-induced liver injury (DILI). Chronic dosing improved DILI detection over short-term cultures, distinguished clinically hepatotoxic drugs from their structurally similar non-toxic analogs, and revealed species differences between human and rat hepatocytes (PMID: 23152190). To assess toxicity and overall cell health, we evaluate endpoints including intracellular ATP and glutathione levels, as well as global gene expression signatures that illuminate the mechanism of action prior to overt injury (PMID: 28369597). The modularity of our platforms further extends these capabilities to more complex biology. Incorporating Kupffer cells, for example, enabled study of cytokine-mediated, inflammation-driven suppression of metabolic enzymes, such as IL-1β– and IL-6–dependent CYP3A4 downregulation, and the resulting disease–drug interactions (PMID: 25739975). Together, these metabolism and toxicity assessments complement disease-efficacy modeling to support the selection of safe, effective therapeutics.

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BIOGRAPHICAL SKETCH

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NAME: Glynis L Kolling

eRA COMMONS USER NAME (credential, e.g., agency login): glk3anih

POSITION TITLE: Associate Professor of Research

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
The Ohio State University	BS	1996	Food Science
Rutgers, The State University of New Jersey	MS	1999	Food Science
Rutgers, The State University of New Jersey	PhD	2003	Food Science
University of Virginia, Charlottesville VA	Post-doc	2003-2005	Nephrology

A. Personal Statement

Dr. Kolling has collaborated closely with Dr. Papin for close to 15 years and brings scientific expertise that bridges her background in microbiology, infectious diseases, and food science to systems level aspects of various projects. She has mentored students and postdocs on numerous project aspects (from the biological side) that examine both bacterial and mammalian physiology and host-microbe interactions. As the chair of the Institutional Biosafety Committee at UVA, Dr. Kolling exercises leadership and communication skills across both research and clinical investigators. She actively participates in all aspects of grant writing, manages regulatory aspects of the laboratory research (e.g., constructing and shepherding IBC, IRB, and ACUC protocols), and is able to plan for long term project goals. Dr. Kolling has worked on several teams composed of clinicians, scientists, and students focused on different aspects of biomedical research. Her knowledge and experience allows her to be well positioned to serve as the project manager for the MOSAIC proposal.

B. Positions, Scientific Appointments, and Honors

2023 – present Associate Professor, UVA Biomedical Engineering
2023 – present Chair of the Institutional Biosafety Committee at UVA
2014 – present Member of the Institutional Biosafety Committee at UVA
2017 – present Assoc. Prof. (courtesy), Medicine – Infectious Disease and Int'l Health
2013-2016 Laboratory Director, UVA Fecal Transplant Program

Honors

2018 Mentor of the Year, UVA School of Engineering – Graduate Student Council
2009 Young Investigator Grant In Probiotics Research
2003-05 Postdoctoral Fellow, NIH Institutional Biodefense Training Grant

C. Contributions to Science

Microbiome-host interactions in health and disease. Bacterial species interact with each other within different host niches to the benefit or detriment of the host through metabolism.

Dillard LR, Glass EM, Kolling GL, Thomas-White K, Wever F, Markowitz R, Lyttle D, Papin JA. Genome-scale metabolic network reconstruction analysis identifies bacterial vaginosis-associated metabolic interactions. Nat Commun. 2025 16(1):4768. PMC12098912.

Medlock GL, Carey MA, McDuffie DG, Mundy MB, Giallourou N, Swann JR, Kolling GL, Papin JA. Inferring Metabolic Mechanisms of Interaction within a Defined Gut Microbiota. Cell Syst. 7(3):245-257.e7. PMC6166237.

Jenior ML, Leslie JL, Kolling GL, Archbald-Pannone L, Powers DA, Petri WA, Papin JA. Systems-ecology designed bacterial consortium protects from severe Clostridioides difficile infection. bioRxiv. 2023 PMC10441344

Experimental systems design for application in computational biology. Defining and building a clinically relevant experimental system that can be physically parsed, sampled, analyzed is paramount to understanding many of the complex biological systems. Furthermore, these systems can be used to test potential therapeutics.

Dougherty BV, Moore CJ, Rawls KD, Jenior ML, Chun B, Nagdas S, Saucerman JJ, Kolling GL, Wallqvist A, Papin JA. Identifying metabolic adaptations characteristic of cardiotoxicity using paired transcriptomics and metabolomics data integrated with a computational model of heart metabolism. PLoS Comput Biol. 2024 20(2):e1011919. PMC10931521.

Glass EM, Dillard LR, Kolling GL, Warren AS, Papin JA. Niche-specific metabolic phenotypes can be used to identify antimicrobial targets in pathogens. PLoS Biol. 2024 22(11):e3002907PMC11611258.

Moutinho TJ Jr, Panagides JC, Biggs MB, Medlock GL, Kolling GL, Papin JA. Novel co-culture plate enables growth dynamic-based assessment of contact-independent microbial interactions. PLoS One. 12(8):e0182163. PMC5540398.

Environment affects bacterial physiology. Bacteria are exposed to metabolic fluctuations in different host niches that affect colonization, antibiotic resistance, virulence factor expression, and ultimately disease.

Powers DA, Jenior ML, Kolling GL, Papin JA. Network analysis of toxin production in Clostridioides difficile identifies key metabolic dependencies. PLoS Comput Biol. (4):e1011076. PMC10166488.

Dunphy LJ, Grimes KL, Wase N, Kolling GL, Papin JA. Untargeted Metabolomics Reveals Species-Specific Metabolite Production and Shared Nutrient Consumption by Pseudomonas aeruginosa and Staphylococcus aureus. mSystems. 2021 6(3):e0048021. PMC8269234.

Kolling GL, Wu M, Warren CA, Durmaz E, Klaenhammer TR, Timko MP, Guerrant RL. Lactic acid production by Streptococcus thermophilus alters Clostridium difficile infection and in vitro Toxin A production. Gut Microbes. 2012 3(6):523-9. PMC3495789.

Link to complete publication list:

https://www.ncbi.nlm.nih.gov/myncbi/1TiQtu7xx5rAC/bibliography/public/

BIOGRAPHICAL SKETCH

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NAME: **Mac Gabhann, Feilim**

eRA COMMONS USER NAME (credential, e.g., agency login): FM9ENIH

POSITION TITLE: Associate Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University College Dublin, Ireland	BENG	05/1997	Chemical Eng.
Johns Hopkins Univ., Baltimore, MD	PHD	05/2007	Biomedical Eng.
Univ. of Virginia, Charlottesville, VA	Postdoc	06/2009	Biomedical Eng.

A. Personal Statement

I have been a faculty member at JHU for sixteen years, am a tenured Associate Professor of Biomedical Engineering, and have published more than 85 papers. My lab is a Computational Systems Pharmacology group studying human disease and therapeutics, using complex mechanistic computational simulations to predict and test new and/or optimized treatments. It is part of the Institute for Computational Medicine, a world's-first institute that applies mathematical modeling to the understanding and treatment of human disease. Our lab works in multiple disease areas including women's health, cancer, peripheral artery disease, and HIV. I have directly mentored six postdoctoral fellows, ten doctoral students, five MS students, and many undergraduate researchers; many trainees have received research awards. I have served in several leadership roles, some of which are described below in *Contributions to Science*.

Ongoing and recently completed projects that I would like to highlight include:

P01AI169768
Li (PI), Role: Core Lead
03/2022-02/2027
High-Definition Characterization of the Persistence and Perturbation of the HIV Reservoir; Mathematical Modeling Core

P01AI131306
Clements (PI), Role: Project Lead
06/2017-05/2024
Modeling HIV Rebound: Role of SIVmac251 Functional Reservoirs and Biomarkers of Reactivation
PROJECT 3: Quantitative Analysis and Computational Modeling of Viral Rebound

R01MH141671
Veenhuis (PI), Role: Co-Investigator
05/2026-01/2031
Elucidating the effects of broadly neutralizing antibody treatment on neuroinflammation and CNS Persistence in SIV-ART macaques

B. Positions, Scientific Appointments, and Honors

2016 -	Associate Professor with tenure, JOHNS HOPKINS UNIVERSITY, Baltimore, MD
2009 - 2016	Assistant Professor, JOHNS HOPKINS UNIVERSITY, Dept. Biomedical Engineering & Institute for Computational Medicine, Baltimore, MD

2012	Arthur C. Guyton Award for Excellence in Integrative Physiology, American Physiological Society (APS)
2012	Sloan Research Fellow, Sloan Foundation
2008	K99/R00 Pathway to Independence Award, NIH/NHLBI

C. Contributions to Science

1. **SYSTEMS PHARMACOLOGY FOR WOMEN'S HEALTH**. We have developed and applied mechanistic models of sex-specific hormones. This includes integrating ODE-based and PDE-based models to quantify the influence of estrogen, progestin, and IL-1b on behaviors of multiple cell types in endometrial tissue. We have also collaborated with scientists and clinicians to build a mechanistic model of sex-specific hormone effects on bone loss, including the impact of allelic variability on the consequences of aromatase inhibitors on bone loss in breast cancer patients, and the potential of drugs to ameliorate this loss.
   a. Mbuguiro W, Holt SE, Griffith LG, Gnecco JS, Mac Gabhann F. Computational modeling of hormone- and cytokine-dependent proliferation of endometrial cells in 3D co-culture. BioRxiv 2025/682837 [Preprint]. March 18, 2026 [cited 2026 May 20]. https://doi.org/10.1101/2025.10.18.682837
2. **SYSTEMS PHARMACOLOGY OF HIV**. We have developed a detailed multiscale model of HIV, from host-pathogen protein interactions to viral and cellular dynamics, and individualize the model with real patient data to understand population variability. We have used the model to study molecular interventions, including targeting the APOBEC3G endogenous antiretroviral proteins and broadly neutralizing antibodies (bNAbs), and to simulate bone marrow transplant and gene therapy as possible HIV-curative methods.
   a. Chang J, Mac Gabhann F. Identifying potential biomarkers of response to CCR5Δ32 HSCT in HIV infection. BioRxiv 2025/646188 [Preprint]. April 3, 2025 [cited 2025 Apr 3]. Available from: https://doi.org/10.1101/2025.03.30.646188
3. **SYSTEMS PHARMACOLOGY OF THE VEGFR SYSTEM**. We have created and validated detailed mechanistic computational models that span from the subcellular (e.g. receptor trafficking and signaling) to the tissue level (e.g. ligand gradients) to the whole body (e.g. compartmental models of pharmacological interventions). We have applied these models of the VEGF-VEGFR system in diseases including peripheral artery disease and various specific forms of cancer. We have used patient data to create populations of individualized models, in order to simulate virtual clinical trials for different therapeutic approaches, including small molecules, antibodies, gene therapy, biomaterials, and cell therapy.
   a. Sarabipour S, Kinghorn K, Quigley KM, Kovacs-Kasa A, Annex BH, Bautch VL, Mac Gabhann F. Impact of ligand binding on VEGFR1, VEGFR2, and NRP1 localization in human endothelial cells. PLOS Comput Biol (2025) 21:e1013254. PMID: 40668864; PMCID: PMC12310042. doi:10.1371/journal.pcbi.1013254.
   b. Ray CMP, Yang H, Spangler JB, Mac Gabhann F. Mechanistic computational modeling of monospecific and bispecific antibodies targeting interleukin-6/8 receptors. PLOS Comput Biol (2024) 20:e1012157. PMID: 38848446 PMCID: PMC1118920 doi:10.1371/journal.pcbi.1012157
4. **ACADEMIC PROGRAM LEADERSHIP.** I have been a leader of academic programs at both the graduate and undergraduate levels: (1) Biomedical Engineering Doctoral Training Program. Admissions Director 2017-2021; Director 2018-2021. (2) Computational Medicine Doctoral Training Program. Associate Director, 2017-2021; Director, 2021-present. (3) Undergraduate Minor in Computational Medicine. Co-developer and co-lead (2014-present). (4) Hopkins Office for Undergraduate Research (HOUR). Founding director (2017-2018).
5. **SCIENCE SERVICE AND OUTREACH.** I have served in many service roles in science, this is a sampling. (1) Co-Editor-in-Chief, PLoS Computational Biology (2019-present). (2) Member, Ethics Subcommittee of the Biomedical Engineering Society (2021-present). (3) Moderator for National Finals, USDOE National Science Bowl (2014-present). (4) openRxiv Scientific and Medical Advisory Board (2025-present).

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BIOGRAPHICAL SKETCH

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NAME: Papin, Jason A.

eRA COMMONS USER NAME (credential, e.g., agency login): papinnih

POSITION TITLE: Harrison Distinguished Teaching Professor of Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of California, San Diego	B.S.	06/2000	Bioengineering
University of California, San Diego	M.S.	12/2002	Bioengineering
University of California, San Diego	Ph.D.	03/2005	Bioengineering

A. Personal Statement

My lab develops methods to quantitatively characterize emergent properties of metabolic networks and applies these systems biology approaches to address biomedical challenges in infectious disease, toxicology, cancer, and other human diseases. My lab has published extensively on the development and use of metabolic network models, high-throughput experimental profiling of metabolism, and the development of computational modeling methods to characterize complex phenotypes. I actively serve in national and international capacities in the systems biology and biomedical engineering research and education communities. My leadership in the international computational biology community (e.g., conference leadership, former Editor-in-Chief of PLOS Computational Biology, former elected member of the Biomedical Engineering Society board) affords me the opportunity to shape key conversations of direct relevance to the field.

For the proposed project, my experience developing computational approaches to integrate and analyze high-throughput data characterizing metabolic networks of the microbiome and human disease will be invaluable to lead the planned efforts. With extensive collaborations among the scientific team, we form a cohesive and experienced group.

Ongoing projects that I would like to highlight include:

NIH – R01-GM147257 Papin (PI) 06/01/23 – 05/31/27
"Systems biology approach to elucidate complex metabolic dependencies in the evolution of antibiotic resistance"
This grant supports the metabolic phenotyping of resistant *Pseudomonas aeruginosa* and *Staphylococcus aureus* and network modeling to predict vulnerabilities that could be targeted to reduce the development of antimicrobial resistance. This proposed R35 grant application will build on this project.

NIH – T32-GM145443 Papin (Contact MPI), Janes (MPI) 07/01/22 – 06/30/27
"Interdisciplinary training in Systems & Biomolecular Data Science"
The major goals of this training grant are to prepare the next generation of systems biologists and biomedical data scientists through coursework and career-development opportunities.

B. Positions, Scientific Appointments, and Honors

2026 Faculty Research Mentor Award, University of Virginia
2022 Research Collaboration Award, University of Virginia
2021-present Harrison Distinguished Teaching Professor of Biomedical Engineering
2020 Team Science Dean's Award for Excellence in Research, U. Virginia School of Medicine
2019-present Cell Systems, Editorial Board
2018 Elected Fellow, Biomedical Engineering Society

2017	Robert J. Kadner Award for Outstanding Graduate Teaching, U. Virginia School of Medicine
2017-present	Professor (courtesy), Biochemistry & Molecular Genetics, University of Virginia
2017-present	Professor (courtesy), Medicine, (Infectious Dis. & Internat. Health), University of Virginia
2016-2025	PLOS Computational Biology, Co-Editor-in-Chief
2016	Elected Fellow, American Institute of Medical and Biological Engineering
2015-present	Professor, Biomedical Engineering, University of Virginia
2014-2019	Molecular Pharmacology, Editorial Board
2012-2016	Standing Member, NIH Biodata Management and Analysis (BDMA) study section
2007	NSF CAREER Awardee
2005-present	Biomedical Engineering Society (BMES) – (2014 – 2017 Elected Board of Directors)
2005-present	International Society for Computational Biology (ISCB)

C. Contributions to Science

Computational tools for metabolic systems biology. My group has pioneered the development of computational tools for the analysis of metabolic networks, including methods for reconciling gene essentiality data; ensemble methods for accounting for uncertainty in the structure of metabolic networks; integration of expression data into metabolic network models to improve predictive value; and automated network reconstruction approaches. These tools are used by groups worldwide.

a. Medlock GL, **JA Papin**+. 2020. Guiding the refinement of biochemical knowledgebases with ensembles of metabolic networks and machine learning. *Cell Systems*. 10(1): 109-119.e3. +Corresponding author. PMCID: PMC6975163.
b. Jenior ML, EM Glass, **JA Papin+**. 2023. Reconstructor: a COBRApy compatible tool for automated genome-scale metabolic network reconstruction with parsimonious flux-based gapfilling. *Bioinformatics*. 39(6): btad367. +Corresponding author. PMCID: PMC10275916.

Metabolic network analyses of microbes for biomedical and industrial applications. My group has published the genome-scale metabolic network reconstructions for microbes with significant health and industrial relevance. Model predictions were validated and drug targets or targets for metabolic engineering applications were delineated. These models have been used extensively by other groups worldwide.

a. Bartell JA, AS Blazier, P Yen, JC Thorgensen, L Jelsback, JB Goldberg, **JA Papin**+. 2017. Reconstruction of the metabolic network of *Pseudomonas aeruginosa* to interrogate virulence factor synthesis. *Nature Communications*. 8: 14631. +Corresponding author. PMCID: PMC5344303.
b. Carey MA, GL Medlock, M Stolarczyk, WA Petri Jr, JL Guler, **JA Papin+**. 2022. Comparative analyses of parasites with a comprehensive database of genome-scale metabolic models. *PLOS Computational Biology*. 18(2): e1009870. +Corresponding author. PMCID: PMC8901074.

Systems analyses of adaptation of host and pathogen during infection. The response of the human host and microbial pathogen during an infection is complex and poorly understood. We have used systems analysis tools (e.g., metabolic network reconstructions, enrichment analysis of expression data) to understand how pathogen and host adapt and respond to each other.

a. Glass EM, LR Dillard, AS Warren, **JA Papin+**. 2024. Niche-specific metabolic phenotypes can be used to identify antimicrobial targets in pathogens. *PLOS Biology*. 22(11): e3002907. +Corresponding author. PMCID: PMC11611258.
b. Dillard LR, EM Glass, GL Kolling, K Thomas-White, F Wever, R Markowitz, D Lyttle, **JA Papin+**. 2025. Genome-scale metabolic network reconstruction analysis identifies bacterial vaginosis-associated metabolic interactions. *Nature Communications*. 16(1):4768. +Corresponding author. PMCID: PMC12098912.

Human disease systems biology. We have used systems biology methods to characterize cellular networks directly coupled with human disease to identify novel drug targets and biomarkers. This work has focused on cancer, heart disease, toxicology, and other diseases with use of the foundational models.

a. Moore CJ, CP Holstege, **JA Papin+**. 2023. Metabolic modeling of sex-specific liver tissue suggests mechanism of differences in toxicological responses. *PLOS Comput Biol*. 19(8): e1010927. +Corresponding author. PMCID: PMC10470949
b. Dougherty BV, CJ Moore, KD Rawls, ML Jenior, B Chun, S Nagdas, JJ Saucerman, GL Kolling, A Wallqvist, **JA Papin+**. 2024. Identifying metabolic adaptations characteristic of cardiotoxicity using paired transcriptomics and metabolomics data integrated with a computational model of heart metabolism. *PLOS Comput Biol*. 20(2): e1011919. +Corresponding author. PMCID: PMC10931521

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Shayn Peirce-Cottler

eRA COMMONS USER NAME (credential, e.g., agency login): SMP6PNIH

POSITION TITLE: Professor of Biomedical Engineering

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
The Johns Hopkins University, Baltimore, MD	B.S.	05/1997	Biomedical Engineering
The University of Virginia	Ph.D.	05/2002	Biomedical Engineering
The University of Virginia	Postdoc	09/2004	Cell Biology, Plastic Surgery, and Biomedical Engineering

A. Personal Statement

My laboratory's core expertise is in developing and validating multi-scale computational models, and deploying them in conjunction with experiments, to study the multi-cell interactions that give rise to structural and functional tissue remodeling over time. With agent-based modeling serving as the backbone of our simulation frameworks, we integrate network models of intracellular signaling and metabolism and models of tissue-level mechanics to span biological phenomena from gene-to-tissue. Our computational models serve as test beds for studying how biochemical and biomechanical signals are transmitted through the environment to drive the individual cell behaviors that lead to tissue-level outcomes. We have used this combined model-experiment approach to study the structural and functional adaptations of the microvasculature during health and disease and to identify new therapeutic approaches for slowing disease progression, restoring homeostasis in diseased tissues, and inducing tissue regeneration. I am a Fellow in the American Institute for Medical and Biological Engineering and the Biomedical Engineering Society and a Past-President of the Microcirculatory Society. I collaborate extensively with clinician scientists and medical doctors, as well as other engineers and biologists, both at the University of Virginia and at other institutions to tackle problems in cardiovascular disease, diabetes, and neurodegeneration.

B. Positions, Scientific Appointments, and Honors

2022-present Department Chair, Dept. of Biomedical Engineering, University of Virginia
2015-present Professor, Dept. of Biomedical Engineering, University of Virginia
2010-2015 Associate Professor, Dept. of Biomedical Engineering, University of Virginia
2004-2010 Assistant Professor, Dept. of Biomedical Engineering, University of Virginia
2002-2004 Postdoctoral Research Associate, Dept. of Biomedical Engineering, University of Virginia
1997-2002 Graduate Research Assistant, Dept. of Biomedical Engineering, University of Virginia

C. Contributions to Science

1- Development of agent-based models to study skeletal muscle regeneration in females.
My research advances sex-specific modeling of skeletal muscle regeneration through integrated computational and experimental approaches (NIH R21AR080415: *"Modeling to design optimized estrogen-specific muscle regeneration treatment",* PMID: 30830675, 38828844, 33970905, 41784392, 40193354). In this work, we developed an estrogen-sensitive agent-based model that captures estrogen-regulated immune dynamics, satellite stem cell behavior, cytokine signaling, and angiogenesis, calibrated using both published data and controlled studies in ovariectomized mice. This framework revealed a nonlinear, dose-dependent role for estrogen, with moderate estrogen levels optimizing regeneration by coordinating macrophage transitions, stem cell expansion, and vascular recovery, while low or high levels impair outcomes. Collectively, these findings establish estrogen as a systems-level regulator of muscle repair and demonstrate that therapeutic efficacy is strongly dependent on hormonal context, motivating sex-specific and hormone-informed treatment strategies.

2- Multimodal analysis of sex- and brain region-specific cell-cell communication in the neurovascular unit (NVU) in Alzheimer's Disease.
My research examines sex- and region-specific vascular dysfunction in Alzheimer's Disease through multimodal data analysis. In this work, we developed a computational framework to analyze brain cell–cell communication using single-nucleus transcriptomic data from post-mortem human brain tissue. Our findings identify pericytes as central regulators of NVU signaling, revealing distinct ligand–receptor interactions and pathway disruptions that vary by sex and brain region. Through both full-network and targeted pairwise analyses, we demonstrate that pericyte-driven signaling programs are selectively rewired in Alzheimer's disease, implicating these cells in regionally localized and sex-biased vascular vulnerability. This pericyte-centric perspective uncovers context-specific reorganization of neurovascular communication networks that cannot be captured by studying individual cell types in isolation. Collectively, this work provides a mechanistic and data-driven foundation for developing targeted, stratified therapeutic strategies that account for sex and brain region in Alzheimer's disease.

3- Multiscale models of mechano-biological regulation of the microvascular adaptations contributing to idiopathic pulmonary fibrosis (IPF).
My research investigates the cellular and molecular origins of fibrotic foci in IPF through an integrated computational–experimental framework (R01HL155143: "*Modeling to Design Treatments for Idiopathic Lung Fibrosis*"; PMID: 32438056, 37248193, 40677170, 39720203, 31920691). In this project, we developed a multiscale agent-based model of pericyte–endothelial interactions that incorporates kinase-mediated signaling pathways (e.g., PDGF, VEGF, TGF-β) and leverages machine learning and Bayesian inference to identify rules governing pericyte-to-myofibroblast differentiation. Guided by murine RNA-seq data and validated in the bleomycin model with lineage-traced pericytes, our work supports the hypothesis that a subset of pericytes detaches from the microvasculature and drives fibrotic foci formation through aberrant signaling. We used the model to simulate responses to FDA-approved therapies (nintedanib, pirfenidone) and novel kinase inhibitors, revealing how spatial heterogeneity and signaling dynamics influence treatment efficacy. By further integrating human single-cell transcriptomic and histopathological data, this framework bridges preclinical and clinical scales to predict patient-specific fibrotic progression and therapeutic response. Collectively, this work provides mechanistic insight into IPF pathogenesis and establishes a predictive platform for designing more effective, spatially and molecularly targeted anti-fibrotic therapies.

OMB No. 0925-0001 and 0925-0002 (Rev. 10/2021 Approved Through 01/31/2026)

BIOGRAPHICAL SKETCH

NAME: Vaughan, Monique Hiersoux

eRA COMMONS USER NAME (credential, e.g., agency login): MHVaughan

POSITION TITLE: Assistant Professor of Obstetrics and Gynecology

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
University of Florida, Gainesville, FL	BS	05/2008	Zoology (summa cum laude)
University of Florida, Gainesville, FL	MD	05/2012	Medicine
University of Virginia, Charlottesville, VA	Resident	06/2016	Obstetrics and Gynecology
Duke University, Durham, NC	Fellow	06/2019	Urogynecology and Reconstructive Pelvic Surgery
University of Virginia, Charlottesville, VA	MS	05/2026	Clinical Research

A. Personal Statement

I am an Assistant Professor and board-certified urogynecologist in the Department of Obstetrics and Gynecology, Division of Urogynecology and Reconstructive Pelvic Surgery at the University of Virginia (UVA) School of Medicine. A large focus of my research portfolio is multidisciplinary and translational collaborations with various aims to understand the pathophysiology of pelvic floor disorders, and to develop novel treatments for these conditions. I have demonstrated success in team science in several areas of translational research, namely understanding the implications of the vaginal and urinary microbiome in disorders of the pelvic floor. I have experience assembling multi-disciplinary teams including clinical colleagues in urogynecology, biomedical and chemical engineers, lab technicians, microbiologists, clinical research coordinators, statisticians and bioinformaticians for various projects. I worked with a team in the clinical microbiology lab during my fellowship to develop a process for performing critical laboratory techniques for female genitourinary (GU) microbiome studies (expanded urine culture). I was awarded institutional grant funding from our department to complete this project, which led to multiple peer-reviewed publications and laid the groundwork for me to continue work in this area and apply for extramural funding. I currently have a prospective study underway to better understand longitudinal changes in the genitourinary microbiome in women using pessaries and have institutional grant funding for the same. Thus, I have the persistence, knowledge and experience to carry out the proposed work as a clinical collaborator for this study. My clinical expertise and prior experience collaborating with multidisciplinary teams are valuable resources to propel the work of the current proposal forward and help develop clinically relevant vaginal microbiome models.

B. Positions, Scientific Appointments, and Honors

2016 - 2019 Clinical Instructor, Duke University Medical Center, Durham, NC
2019 - Assistant Professor, University of Virginia School of Medicine, Charlottesville, VA

Honors (most notable)

2011 Chapman Chapter of the Gold Humanism Honor Society, UF College of Medicine
2014 Alpha Omega Alpha (AOA) Honor Medical Society, University of Virginia
2016 Anne L Brodie Resident Clinician Award, University of Virginia
2016 Best Resident Research Project, UVA Department of OB/GYN Research Day
2017 Winner of Clinical Poster Session, Duke Multidisciplinary Benign Urology Research Day
2021 APGO Excellence in Teaching Award, University of Virginia School of Medicine
2022 Academy for Excellence in Education, University of Virginia School of Medicine
2024 American College of Obstetricians and Gynecologists (ACOG) & Council on Resident Education in OBGYN (CREOG) National Faculty Award for Excellence in Resident Education

C. Contributions to Science

1. **Genitourinary microbiome:** With a team of multidisciplinary collaborators, I researched alterations in the urinary microbiota and microbiome in women with recurrent urinary tract infections and highlighted key differences in the urinary microbiome in healthy menopausal women compared to menopausal women with recurrent urinary tract infections. We also published a novel manuscript evaluating the concordance between two common laboratory techniques for characterizing the urinary microbiome, expanded quantitative urine culture (EQUC) and 16s amplicon sequencing. These publications had significant impact and laid the groundwork for further study. My recent work also includes an analysis of longitudinal changes in the vaginal and urinary microenvironments in women using pessaries for which I have institutional grant funding for preliminary data that will be utilized for follow-on extramural funding.
   a. **Vaughan, M. H**., Mao, J., Karstens, L. A., Ma, L., Amundsen, C. L., Schmader, K. E., & Siddiqui, N. Y. (2021). The Urinary Microbiome in Postmenopausal Women with Recurrent Urinary Tract Infections. The Journal of urology, 206(5), 1222–1231.
   b. **Vaughan MH**, Gregory E. Zemtsov, Erin M. Dahl, Lisa Karstens, Li Ma, Nazema Y. Siddiqui. "Concordance of Urinary Microbiota Detected by 16S rRNA Amplicon Sequencing Versus Expanded Quantitative Urine Culture." Am J Obstet Gynecol. 2022 Jun 25:S0002-9378(22)00485-9. doi: 10.1016/j.ajog.2022.06.031. PMID 35764138
   c. **Grant funding:** "Longitudinal assessment of the vaginal microbiome in women using pessaries." Ferguson Faculty Research Award, University of Virginia Department of Obstetrics and Gynecology. Total Awarded: $40,000. Role: Principal Investigator. Timeline: 2025/03/01-2027/06/30

2. **Urinary tract infections:** Recurrent UTIs are highly bothersome to patients and are associated with decreased quality of life. Despite appropriate prophylaxis with first (vaginal estrogen) and second line agents (methenamine, D-mannose, cranberry tablets, etc), many women continue to struggle with UTI recurrence. My colleagues and I studied how the urinary microbiome may be implicated in menopausal women with recurrent urinary tract infections, and we were also interested in comparing various antibiotic regimens that may be used to prevent UTI at the time of intravesical Botox injections. Studies associated with these questions are listed below.
   a. Bickhaus JA, **Vaughan MH**, Truong T, Li Y, Siddiqui NY. A comparison of antibiotic prophylaxis regimens to decrease the risk of post-procedure urinary tract infection after onabotulinum toxin A injection. *Int Urogynecol J*. 2020 Jan 27. doi: 10.1007/s00192-020-04230-7. PMID: 31989204
   b. **Vaughan MH,** Mao J, Karstens LA, et al. The Urinary Microbiome in Postmenopausal Women with Recurrent Urinary Tract Infections. J Urol 2021;206:1222-31.

3. **Fibrous hydrogels to augment pelvic organ prolapse repair**: I have worked with a team of chemical and biomedical engineers at UVA to develop and study the safety and impact of a supramolecular fibrous hyaluronic acid hydrogel to augment uterosacral ligament suspensions (USLS). The development of novel, non-mesh materials to augment native tissue prolapse repairs is critical, as we work in the field of urogynecology to balance the superior efficacy but higher complication rates of polypropylene mesh-based prolapse repairs. Through this collaboration, our team developed the hyaluronic acid fibrous hydrogel as well as the novel rat animal model to study the hydrogel-augmented USLS. This collaboration led to publications, patents and grant funding. This work is ongoing and we are applying for R01 funding.
   a. Miller BJ, Jones BK, Turner JS, Caliari SR, **Vaughan MH**. Development of a uterosacral ligament suspension rat model. J. Vis. Exp 2022 Aug 17;(186). doi: 10.3791/64311. PMID: 36063010
   b. B. Miller, W. Wolfe, J.L. Gentry, M.G. Grewal, C.B. Highley, R. De Vita, **M.H. Vaughan****, S.R. Caliari**, 'Supramolecular fibrous hydrogel augmentation of uterosacral ligament suspension for treatment of pelvic organ prolapse,' *Advanced Healthcare Materials*, 522930, May 2023. **co-corresponding authors.
   c. **Patent:** S.R. Caliari, B. Miller, **M. Vaughan**, *In vivo augmentation of uterosacral ligament suspension with a supramolecular fibrous hydrogel for treatment of pelvic organ prolapse*, US Provisional Patent Application 63/426,466, November 18, 2022. Full patent application 3062/194 PCT, November 20, 2023.

Complete List of Published Work:
https://pubmed.ncbi.nlm.nih.gov/?term=Vaughan,Monique%5BAuthor%5D

OMB No. 0925-0001 and 0925-0002 (Rev. 10/2021 Approved Through 01/31/2026)

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Vincent Bensan Young

eRA COMMONS USER NAME (credential, e.g., agency login): YOUNGVI

POSITION TITLE: Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE *(if applicable)*	Completion Date MM/YYYY	FIELD OF STUDY
MIT, Cambridge, MA	postdoctoral	Sept 2000	Microbiology
MGH, Boston, MA	resident	Jun 1998	Internal Medicine/ID
Stanford University, Stanford, CA	MD	Jun 1992	Medicine
Stanford University, Stanford, CA	PhD	Jun 1992	Micro/Immuno
MIT, Cambridge, MA	BS	Jun 1985	Life Sciences

A. Personal Statement

My training as a biomedical researcher and infectious diseases physician positions me to serve as an investigator on this project that will develop a multi-scale computational model to understand the role of sex-specific differences that lead to differential responses to therapies. For over 25 years I have studied classic molecular pathogenesis using in vitro and animal models of disease. I also conduct translational research on bacterial infectious diseases. Our group has considerable experience in the analysis of microbiome data from multiple body sites including the gastrointestinal tract, (PMC11841841), respiratory tract (PMC4358017) and female reproductive tract (PMC12993637, PMC4626584). This work has led to over a hundred peer review publications in these areas. We are well positioned to carry out the proposed work. I recently led an NIH-funded R01 grant (AI162787) that examined how the gastrointestinal microbiota varied between male and females of the full age range using murine models. This work revealed distinct differences between the microbiota of male and female mice and that these differences increased with age, leading to differential responses to experimental infection with the nosocomial pathogen *Clostridioides difficile* (PMC1268656). I have worked closely with researchers with expertise in epidemiology, bioinformatics and modeling so I look forward to being part of the interdisciplinary team being assembled by Dr. Papin.

B. Positions, Scientific Appointments, and Honors

2016 - Professor, University of Michigan, Department of Internal Medicine, Infectious Diseases Division and Microbiology & Immunology, Ann Arbor, MI
2010 - 2016 Associate Professor, University of Michigan, Department of Internal Medicine, Infectious Diseases Division and Microbiology & Immunology, Ann Arbor, MI
2007 - 2010 Assistant Professor, University of Michigan, Department of Internal Medicine, Infectious Diseases Division and Microbiology & Immunology, Ann Arbor, MI
2000 - 2007 Assistant Professor, Michigan State University, College of Human Medicine, Department of Microbiology & Molecular Genetics, National Food Safety & Toxicology Center, East Lansing, MI
1999 - 2000 Instructor, Harvard University, Medical School, Boston, MA
1999 - 2000 Clinical Assistant, Harvard University, Harvard Medical School, Boston, MA
1998 - 2000 Research Fellow, Massachusetts General Hospital, Infectious Diseases, Boston, MA

1998 - 2000 Graduate Assistant, Massachusetts General Hospital, Infectious Diseases, Boston, MA
1996 - 2000 Postdoctoral Fellow, Massachusetts Institute of Technology, Department of Bioengineering & Environmental Health, Cambridge, MA
1995 - 1996 Clinical Fellow, Massachusetts General Hospital, Internal Medicine/Infectious Diseases, Boston, MA
1995 - 2001 Associate Staff Physician, Massachusetts Institute of Technology, Medical Department, Cambridge, MA
1992 - 1995 Resident physician, Massachusetts General Hospital, Department of Internal Medicine, Boston, MA

C. Contributions to Science

1. Understanding the role of microbes in human health and disease. My initial training in microbiology/infectious diseases provided me with expertise in bacterial microbial pathogenesis. I have combined this training with the microbial ecology expertise I obtained early in my independent research career. While the bulk of my work has been focused on the gastrointestinal tract, I have experience in studying the role of complex microbial communities and pathogens in the upper respiratory tract, the lungs and lower respiratory tract, and the genitourinary tract, including the vaginal tract. I have leveraged my training as an infectious diseases physician to understand that these microbial communities do not exist in isolation, but can influence host physiology through their complex interactions that also involves host immune responses.

2. Applications of microbial ecology to gastrointestinal disease. My early independent research career was characterized by exploring how the study of microbial communities could impact the study of disease within the GI tract including antibiotic-associated diarrhea, the inflammatory bowel diseases and infectious colitis. This research began well before the Human Microbiome Project and afforded me the ability to carefully test how knowledge of the structure and function of the gut microbiota could impact host health and disease. I have leveraged my training in bacterial pathogenesis, microbial ecology and clinical infectious diseases to employ a systems approach to understanding complex host-microbe interactions. I have employed advanced in vitro model systems (bioreactors and organdies), murine models and studies on human subjects to understand these host-microbe interactions at multiple spatial and temporal scales.

3. Leading and contributing to teams of scientists in collaborative, interdisciplinary approaches to complex biomedical questions. Historically, the “team science” approach to biomedical research was not common. However, the bulk of my career has involved working, both as a leader and a member in large teams of scientists to address complex questions including the study of cystic fibrosis, inflammatory bowel disease, *C. difficile* infection, antibiotic resistance and foodborne illness. I have come to appreciate that although there can be difficulties in maintaining the team approach, I enjoy the opportunity to work together, combining various interests and expertise to push the scientific envelope.

4. Translational study of the epidemiology and pathogenesis of healthcare-associated infections. As an infectious diseases physician and a microbiologist, I have blended these aspects of my training and interest to translate findings from the bench to the bedside and back again. This work has involved working with diverse teams of investigators to tackle complex problems using a panel of approaches.

Equipment and Facilities

Environment**:** All the investigators have working relationships with other investigators on the team through previously funded or proposed projects. The established relationships have the potential to increase the project momentum with greater sharing and utilization of unique resources among the institutions summarized below. Across the MOSAIC team, ample equipment (standard and specialized) and core facilities are available to carry out experiments and analyses as described in the research strategy. All the universities are equipped with various high performance computing equipment that will be essential for the more complex steps of model construction, analysis, and integration and are detailed below. We will leverage the use of the most cutting-edge technologies available across the team to generate high quality data as necessary.

Computing infrastructure**:** The distributed computational infrastructure across participating institutions provides both the capacity and redundancy necessary for model development, validation, reproducibility testing, and long-term maintenance of the MOSAIC software ecosystem. All lab personnel under the multiple investigators are provided with at least one laptop with additional workstations available that may be dedicated to strict analysis or to software associated with an instrument. Various software packages are provided through the universities for data/statistical analysis, word processing, and figure generation. Investigators will make use of HPC for their respective parts on the project and resulting data and code will be shared through a common MOSAIC repositories and institutional storage infrastructure as appropriate.

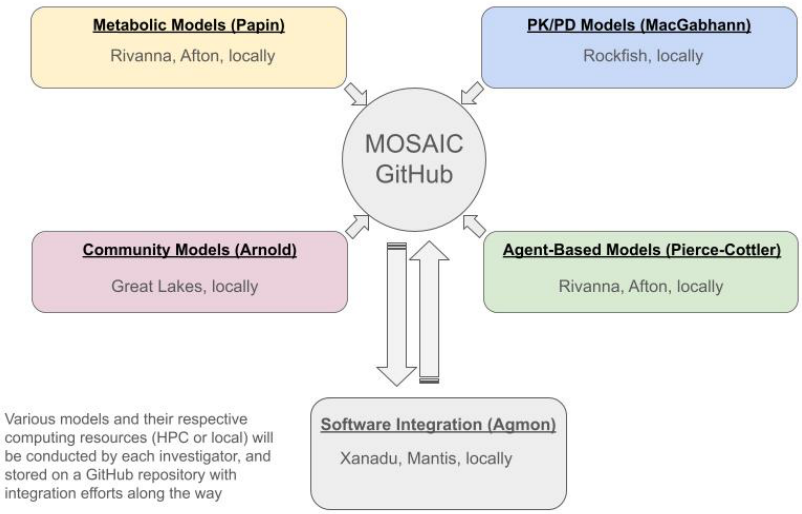


Figure 1. HPC resources across MOSAIC partners

High-performance computing clusters at the **University of Michigan** (UM) (Arnold, Young) include those at the Center for Computational Medicine and Bioinformatics (CCMB), which supports a computing infrastructure that consists of a number of computing clusters and independent computing and application servers all connected to a high-speed campus network. ***Great Lakes*** is an HPC Linux-based cluster intended to support parallel and other applications that are not suitable for departmental or individual computers. Each *Great Lakes* compute node comprises multiple CPU cores with at least 4 GB of RAM per core; *Great Lakes* has approximately 14,000 cores. All compute nodes are interconnected with InfiniBand networking. The larger memory *Great Lakes* hardware comprises 3 compute nodes, each configured with 1.5 TB RAM. *Great Lakes* contains 20 GPU nodes, with a total of 40 NVIDIA Tesla V100 CUDA capable GPUs. There are also 4 visualization nodes available, each equipped with a single NVIDIA Tesla P40. Computing jobs on *Great Lakes* are managed through the SLURM Scheduler. These servers have access to storage systems that collectively total greater than 200 TB of storage. Battery backup, fault monitoring, data snapshots, site-distinct mirroring and off-site backups are part of our comprehensive backup/recovery strategy. CCMB has a large and diverse research computing base, including the National Center for Integrative Biomedical Informatics (NCIBI), **one of seven** NIH-funded National Centers for Biomedical Computing; the Molecular and Behavioral Neuroscience Institute (MBNI); and the Medical

School Information Systems (MSIS). MSIS provides servers, virtual servers, and storage and networking connectivity for the UMHS backbone. CCMB and MSIS together have developed research IT support for the UM Sequencing Core especially for next-generation sequencing technologies and data analysis needs; with access to an 80-core computer cluster with over 100 TB of dedicated storage. NCIBI/CCMB computational infrastructure consists of database servers, computer servers, virtual servers, storage backup and recovery, and staff. The computing sites are connected by a campus network backbone consisting of a mixture of Gigabyte and increasingly 10-Gigabyte Ethernet. The campus network is connected to the Internet and Internet2. University of Michigan researchers and CCMB are also affiliated with several computing consortia, such as computational infrastructure initiatives.

Research Computing at the **University of Virginia** (UVA) (Papin, Pierce-Cottler) is a centralized resource providing high performance computing and hundreds of pre-installed software packages available for computational research across many disciplines. The High-Performance Computing systems ***Afton*** and ***Rivanna*** are available to all UVA faculty, staff, students, and postdoctoral associates. The *Afton* HPC system consists of 300 computer nodes each with 96 compute cores based on the AMD EPYC 9454 architecture for a total of 28,800 cores. Each *Afton* node boasts a minimum of 750 GB of memory per node. The *Rivanna* HPC system has 603 nodes with over 20,476 cores and significant storage. 7 NVIDIA HGX B200 GPU nodes will be added to the *Afton* high-performance computing (HPC) cluster this summer—a major leap forward for AI and HPC workloads. Built on NVIDIA's Blackwell architecture, B200 systems are optimized for large-scale AI training and inference as well as data-intensive HPC applications. Each node provides 2TB of node CPU memory and 8-way connected Tensor Core GPUs with 180GB of VRAM memory. These devices are designed to support trillion-parameter AI models, AI-for-science workloads, advanced simulations, and exascale-class computing environments

Shared Computational Facilities at the ARCH (Advanced Research Computing at **Johns Hopkins University** (Mac Gabhann)) facility administers state of the art high performance computing resources, manages highly reliable data storage, and provides outstanding collaborative scientific support to empower computational research, scholarship, and innovation. ARCH is located on the Bayview Campus of Johns Hopkins University and funded by a State of Maryland grant to Johns Hopkins University through the Institute for Data Intensive Engineering and Science (IDIES). The main cluster administered by ARCH, ***Rockfish***, has 34,128 cores (711 nodes), a combined theoretical performance of 3.3 PFLOPs and Rmax of 2.1 PFLOPs. Rockfish has three parallel file systems (GPFS) with a total of ~13PB of usable space. It employs Mellanox Infinidad HDR100 connectivity (1:1.5 topology). All JHU PIs are granted a startup allocation on *Rockfish* (50,000 hours) and can then apply for project-specific access of specific resources at any time at no cost. *Rockfish* follows the “condominium model” with three main units. The first unit is based on an NSF Major Research Infrastructure grant, a second unit contains mainly medium-sized condos (for example DURIP/DoD, Deans’ contributions condos), and the last unit is a collection of individual research groups condos. All three units are shared, with no physical separation, by all users. PIs can use their own funding to integrate resources (condos) to the main cluster. Condos may consist of compute nodes, GPU nodes, sub-racks, management and Infiniband switches, and any necessary software licenses.

The Center for Cell Analysis and Modeling at the **University of Connecticut** (UConn) (Agmon) maintains and provides full free access to the High-Performance Computing Facility (HPC) that includes a ca.1,000 sq. ft. datacenter with a Internet2 connection, and a 750 sq. ft. software development suite. The ***Xanadu, Mantis*** and ***NMR*** clusters comprise over 12,000 CPU Cores and 77 TB of RAM capable of processing at 360 TFLOPs. Over 22 PB of storage including 775 TB Quantum, 3.4 PB EMC Isilon, 10.5 PB Qumulo scale-out clusters, 2 PiB Spectra tape backup and archive along with 2.3 PB Amplidata, and 3.1 PB Scality Geo-Spread cloud storage is part of the

HPC. CCAM's successful strategy for dissemination of technology developed through a R24 grant is a model for dissemination of science and technology for other projects at the Center. Strong ties also exist between CCAM and the recently established Institute for Systems Genomics, the Center for Quantitative Medicine, and the structural biology group within the Department of Molecular Biology and Biophysics all of which utilize the CCAM HPC facility.

Clinical resources**:** Several of the universities participating in this proposal are affiliated with health systems that provide patient care for the surrounding areas. These clinical resources provide access to the biospecimens, patient-derived data, and associated metadata necessary for model calibration, validation, and hypothesis generation in BV and MASLD. Two of the MOSAIC investigators (Vaughan, Young) have clinical appointments and are therefore well positioned to assist with clinical data. Both **UVA** and the **UM** are multi-specialty academic medical centers and research institutions for clinical, translational, and basic science investigations. Collection of clinical specimens and bacterial isolates for IRB-HSR approved research have been conducted to understand the clinical patient landscape. A number of investigators maintain laboratories that are approved through their local institutional biosafety to handle BSL2 level material (bacteria and human derived materials)

Aggregated and other clinical datasets are available through UVA's **Resource Data Exchange (RDE)** and also the **Research Data Warehouse (RDW)** at UM. The RDE connects to IRBs for HIPAA-protected data, supports auto-extracts from REDCap, and offers optional automated DICOM deidentification. Currently, there are over 3,000 resources and 1,700 datasets available through the RDE. The RDW is a SQL Server relational database with either nightly, weekly, or real time extracts from more than 10 distinct data sources across the enterprise. Currently, there are more than 3,800,000 unique patients and 92,000,000 encounters in the RDW. Data sources include patient demographics, labs, medications, diagnoses, procedures, encounters/visits, and data about available biospecimens in a central biorepository. These data are for research purposes available campus wide.

Common Equipment and Cores**:** Laboratories capable of conducting "benchtop" experiments (Arnold, Derre, Khetani, Papin, Pierce-Cottler, Young) have standard equipment for basic molecular biology (isolation and detection), eukaryotic cell culture (growth, visualization, detection), and prokaryotic culture (growth and detection) within 1,000-2,000 ft^2 of lab space. These labs all operate under approved BSL-2 practices to contain known and adventitious pathogens that may be contained in human derived material. In addition to common equipment, core facilities are available at all of the institutions to handle sample processing and downstream analysis using advanced 'omics techniques (genomics [whole genome and metagenomics], transcriptomics, targeted/untargeted metabolomics with liquid handling), advanced microscopy (cryo- and electron microscopy), bioinformatics, and biorepositories/storage facilities. All cores operate on a fee-for-service and in many cases have partial subsidies for university-based investigators.

Specialized "benchtop" related equipment**:** Advanced culture techniques and detection for organisms that require microaerophilic or anaerobic conditions are readily conducted in hard and soft-sided culture chambers (Papin, Young). Bioprinters for advanced cell culture configurations (tissue-on-a-chip) are available (Khetani, Pierce-Cottler) along with visualization using confocal and fluorescence microscopy capable of live cell imaging with appropriate analytic software (Derre, Khetani, Pierce-Cottler).

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Resource Sharing Plan

The MOSAIC team is committed to rapid dissemination of models, software, datasets, and documentation to maximize transparency, reproducibility, and community reuse. Importantly, during the model integration phase, we anticipate having several people across the modeling efforts that can cross-check/validate code function. A MOSAIC GitHub will be used for active development, version control, issue tracking, collaborative development, and dissemination (see "*Computing Infrastructure*" in the Equipment and Facilities Section and "*Public repositories, Software and Licensing, and Preservation and Timelines"* in the Data Management/Sharing Section).

Model Components and Modeling Parameters: The computational models/modeling components will be made available through community repositories as available (e.g., Biomodels at EBI). The standard format for such computational models is SBML (systems biology markup language). Models will also be made freely available through public GitHub repositories, as well as through supplementary material in accompanying publications. Documentation detailing model specification, parameter sets, simulation configuration/results, metadata, and other datasets will be freely available through project repositories and associated documentation sites when appropriate. In addition, software and algorithms that are developed during this project will be made available with appropriate repositories (e.g., GitHub). We will similarly make newly developed computer code available. Resources not suitable for repository deposition will be distributed upon reasonable request, consistent with institutional, ethical, and regulatory requirements. The investigators have a strong track record of publicly disseminating computational models and software. For example, Papin and Mac Gabhann co-authored the code availability policy of PLOS Computational Biology {PMID 35653366} which has been championed across the publishing industry.

Associated Scientific Data: The use of publicly available data will be cited with the corresponding models and include all relevant metadata. Any nucleic acid sequence read data generated uniquely through this project will be annotated and deposited to NCBI *Sequence Read Archive*. High-throughput metabolomic data will be uploaded to *MetaboLights*, an open-access metabolomic data repository by European Bioinformatics Institute.

In the case where samples may be shared as dictated by approved IRB protocols, material availability, and in accordance with the proposed work, we would follow all shipping regulations and use the Uniform Biological Material Transfer Agreement and/or consult with University counsel regarding any materials transferred.

Documentation and reproducibility: The MOSAIC team members have championed the reproducibility of computational models {PMID 32845080, 32427998, 41343683}. We will implement practices to ensure that there is sufficient documentation and metadata to enable model reproduction and independent validation. All code development will occur with field standards in version control and collaborative coding efforts. By embedding personnel from each modeling-focused lab across the BV, MASLD, Integration, and Software teams, we will promote cross-laboratory awareness, compatibility, and early identification of reproducibility issues. All software and computer models will be checked by the Software team (Agmon) for criteria of proper documentation and delineation of software dependencies. Agmon was a member of the Center for Reproducible Biomedical Modeling {PMID 41343683} and he will ensure that MOSAIC is implementing best practices.

Timelines: Curated datasets, models, software releases, and documentation supporting publications will be publicly released at the time of publication, with interim releases made whenever feasible to accelerate community use and feedback. GitHub will be used for active development, while stable releases and associated datasets may be archived through Zenodo or similar long-term repositories with persistent identifiers. We will not delay disclosure of research findings for the purpose of evaluating an invention for a patent application or the actual filing of a patent application.

Open model infrastructure and sharing: To make MOSAIC more than a one-off integrated model, we will build it around open, modular software infrastructure for composing, sharing, and coordinating model development across teams. Each model component will have an explicit interface that defines its inputs, outputs, assumptions, units, timescale, and biological scope, allowing microbiome dynamics, metabolic models, agent-based models, and whole-body compartment models to be developed and validated independently while still working together in larger simulations. Consistent with the goals of OTA-26-004, MOSAIC models, software, and supporting resources will be designed to enable reuse, extension, and integration by external investigators studying related hormone-regulated biological systems. Internal project collaboration will use institutionally approved file-sharing platforms. Public dissemination will occur through established repositories appropriate to the resource type. We will use GitHub-based infrastructure for version control, issue tracking, documentation, code review, and coordination across modeling teams, and distribute reusable software components through PyPI where appropriate. Models, parameter sets, simulation workflows, and validation datasets will be packaged with clear provenance and documentation so they can be reused, extended, and compared across teams. Sharing of human-subjects data will comply with IRB approvals, informed consent provisions, HIPAA requirements, and institutional policies governing protected health information. This open infrastructure will make MOSAIC more reproducible, easier to validate, and adaptable beyond the initial use cases.

Data Management and Sharing Plan

Data Types: The MOSAIC project will generate scientific data, software, documentation, and training materials related to various types of computational modeling and simulation across the investigators. Expected computation-related outputs include model specifications, simulation configurations, parameter sets, structured simulation results, metadata, schemas, analysis scripts, notebooks, APIs, source code, documentation, tutorials, and example datasets. Expected biological data related outputs include genomics, 16S sequencing, transcriptomics, metabolomics, and other cell based measures.

Reproducibility and Validation: Sharing these outputs will support transparency, reproducibility, independent validation, and reuse by the broader modeling and scientific communities. Metadata and documentation will describe data formats, simulation parameters, model assumptions, workflows, and methods needed to reproduce published results. Any specialized tools, software, or code needed to access or manipulate the shared scientific data will be readily provided through the computational model repositories or supplemental files of manuscripts with links to community tools that are freely accessible. Validation datasets, benchmark results, and associated metadata used to assess model performance will be shared whenever possible to facilitate independent evaluation of model credibility.

Model and Simulation Standards: Model-related outputs will be shared using community standards whenever possible. Models will be represented in SBML where appropriate, including relevant parameters, environmental conditions, and simulation assumptions. Simulation metadata, experimental designs, and workflow specifications will be described using SED2, a community standard for describing simulation experiments and workflows, when applicable.

Public Repositories: Curated models suitable for broader reuse will be deposited in public repositories such as BioModels, and published simulation results may be shared through BioSimulations when appropriate. GitHub will be used for active development, version control, issue tracking, collaborative development, and dissemination. Any nucleic acid sequence read data generated uniquely through this project will be annotated and deposited to NCBI *Sequence Read Archive*. High-throughput metabolomic data will be uploaded to *MetaboLights*, an open-access metabolomic data repository by European Bioinformatics Institute. The metadata will also be linked as appropriate by filename to the figures of the associated preprints/publications/repository files.

Software and Licensing: Custom software, schemas, workflows, and analysis tools will be released through public GitHub repositories under permissive open-source licenses such as Apache 2.0, MIT, or BSD, unless project-specific constraints require otherwise. Repositories will include README files, license information, citation guidance, installation instructions, examples, tests where feasible, and documentation for both users and developers.

Formats and FAIR Principles: All labs will prioritize open, machine-readable, and interoperable formats, including SBML, SED2, JSON, YAML, CSV, HDF5, and other community-supported formats as appropriate. The MOSAIC team will follow open-science and FAIR principles to maximize findability, accessibility, interoperability, and reuse.

Preservation and Timelines: Curated datasets, models, software releases, and documentation supporting publications will be made publicly available as soon as possible. GitHub will be used for active development, while stable releases and associated datasets may be archived through Zenodo or similar long-term repositories with persistent identifiers. All scientific data generated (experimental, observational, derived from public repositories, simulations) will be made publicly available before a formal patent application is filed. Data sharing will not be delayed for the purpose of evaluating an invention for patent application or filing a patent application.